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Berger et al.

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(54) **INTEGRATED SHOWERHEAD**

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H05K 2201/09063; H05K 2201/10083;

H05K 2201/10522; H05K 2201/10545

See application file for complete search history.

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C23C 16/455 (2006.01)

C23C 16/50 (2006.01)

H05K 1/02 (2006.01)

H05K 1/18 (2006.01)

(52) U.S. Cl.

(57)

ABSTRACT

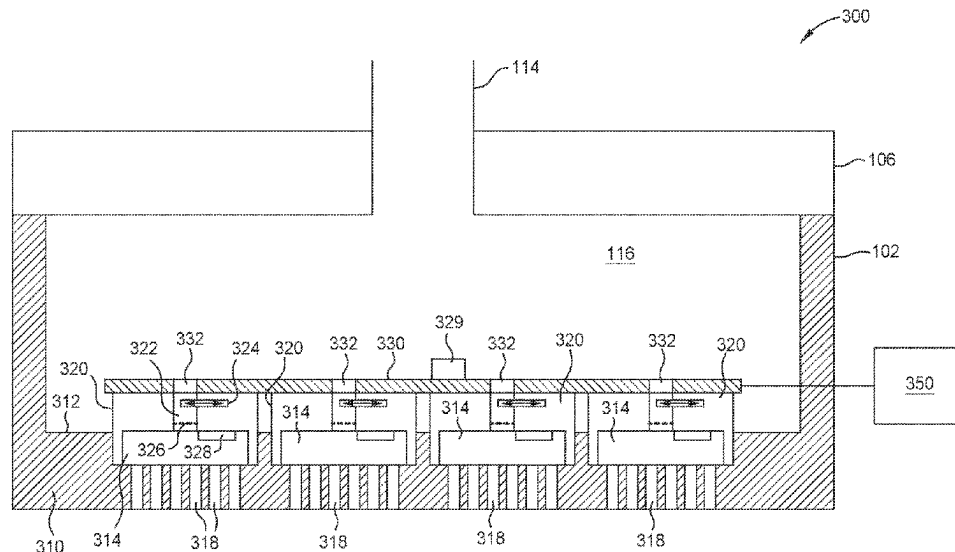
CPC .. *H01J 37/32449* (2013.01); *C23C 16/45565*
(2013.01); *C23C 16/45587* (2013.01); *C23C*
16/50 (2013.01); *H05K 1/0272* (2013.01);
H05K 1/181 (2013.01); *H05K 2201/09063*
(2013.01); *H05K 2201/10083* (2013.01); *H05K*
2201/10522 (2013.01); *H05K 2201/10545*
(2013.01)

A showerhead for a processing chamber includes a faceplate with a plurality of openings. A plurality of compartments are recessed into a top surface of the faceplate. The showerhead includes a plurality of MEMS devices. Each MEMS device is disposed in a corresponding compartment of the plurality of compartments. A printed circuit board including a plurality of ports therethrough is coupled to each MEMS device. Each MEMS device is configured to regulate a gas flow into each corresponding compartment through a corresponding port of the plurality of ports in the printed circuit board.

(58) **Field of Classification Search**

CPC C23C 16/4404; C23C 16/45502; C23C
16/45565; C23C 16/45587; C23C
16/45589; C23C 16/50; C23C 16/52;
H01J 37/3244; H01J 37/32449; H01L

20 Claims, 14 Drawing Sheets



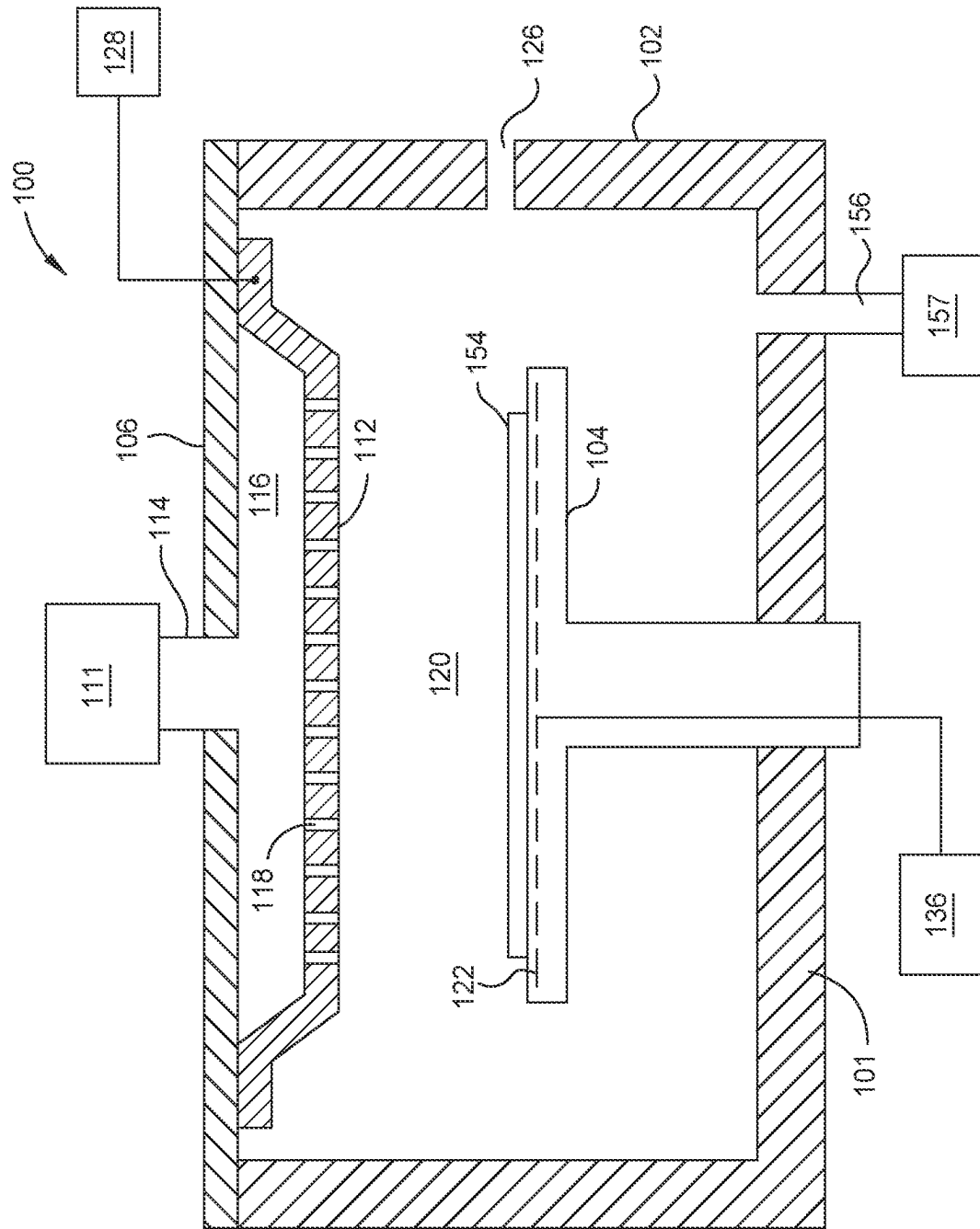
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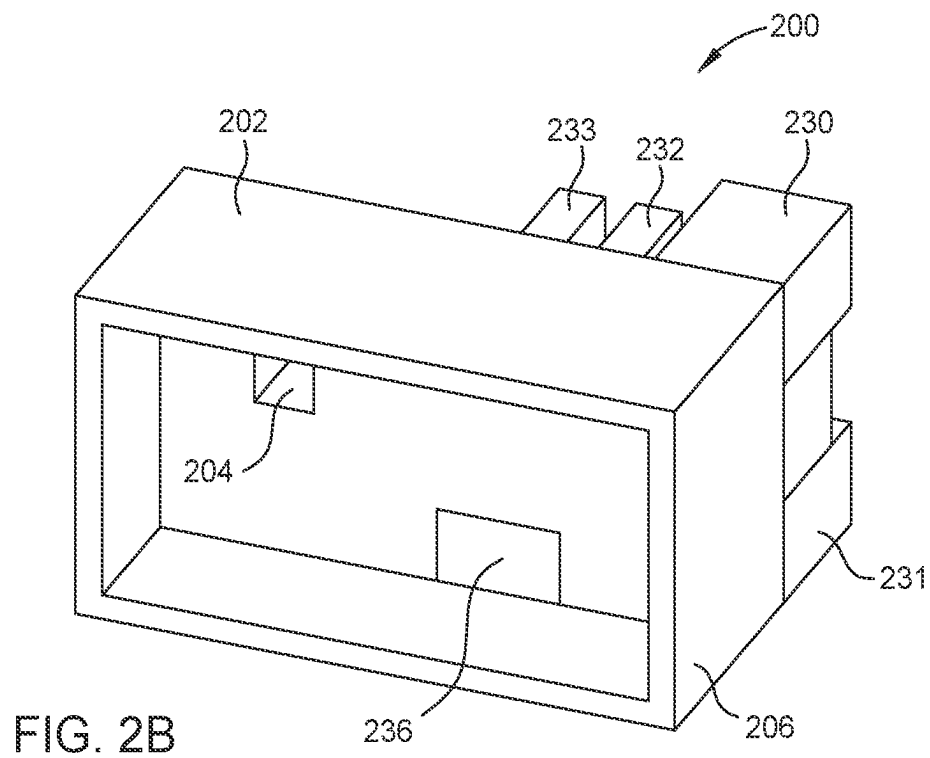
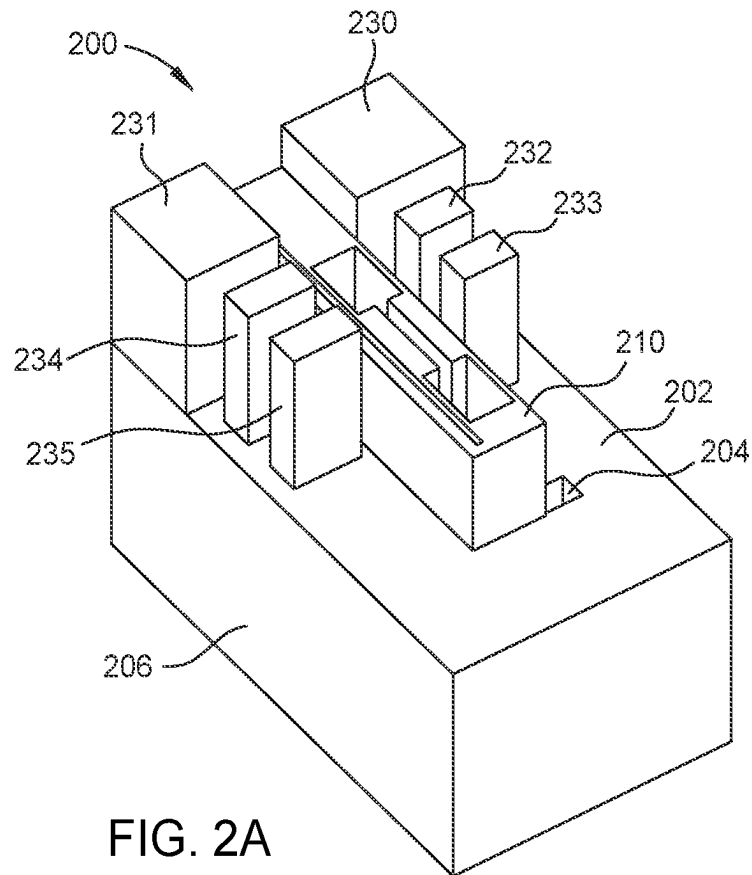
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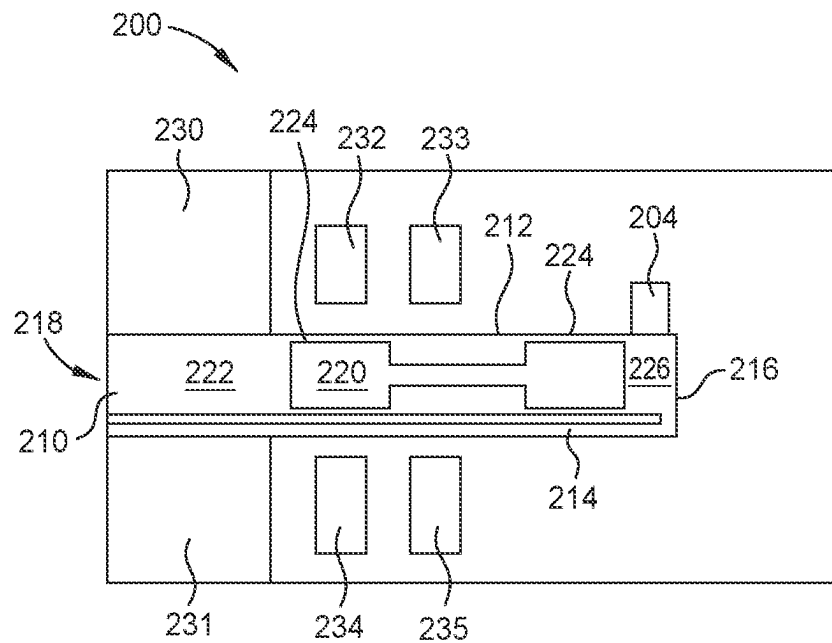


FIG. 2C

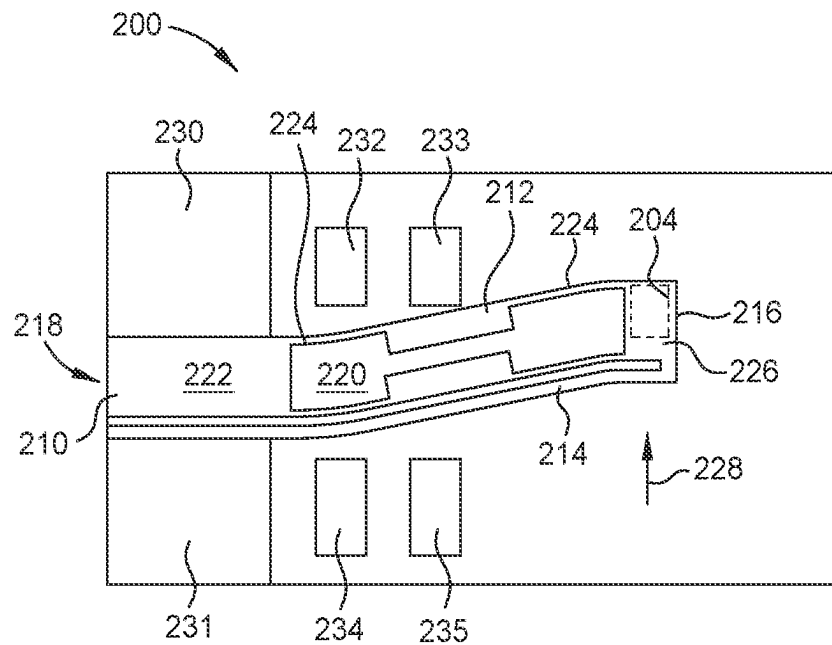


FIG. 2D

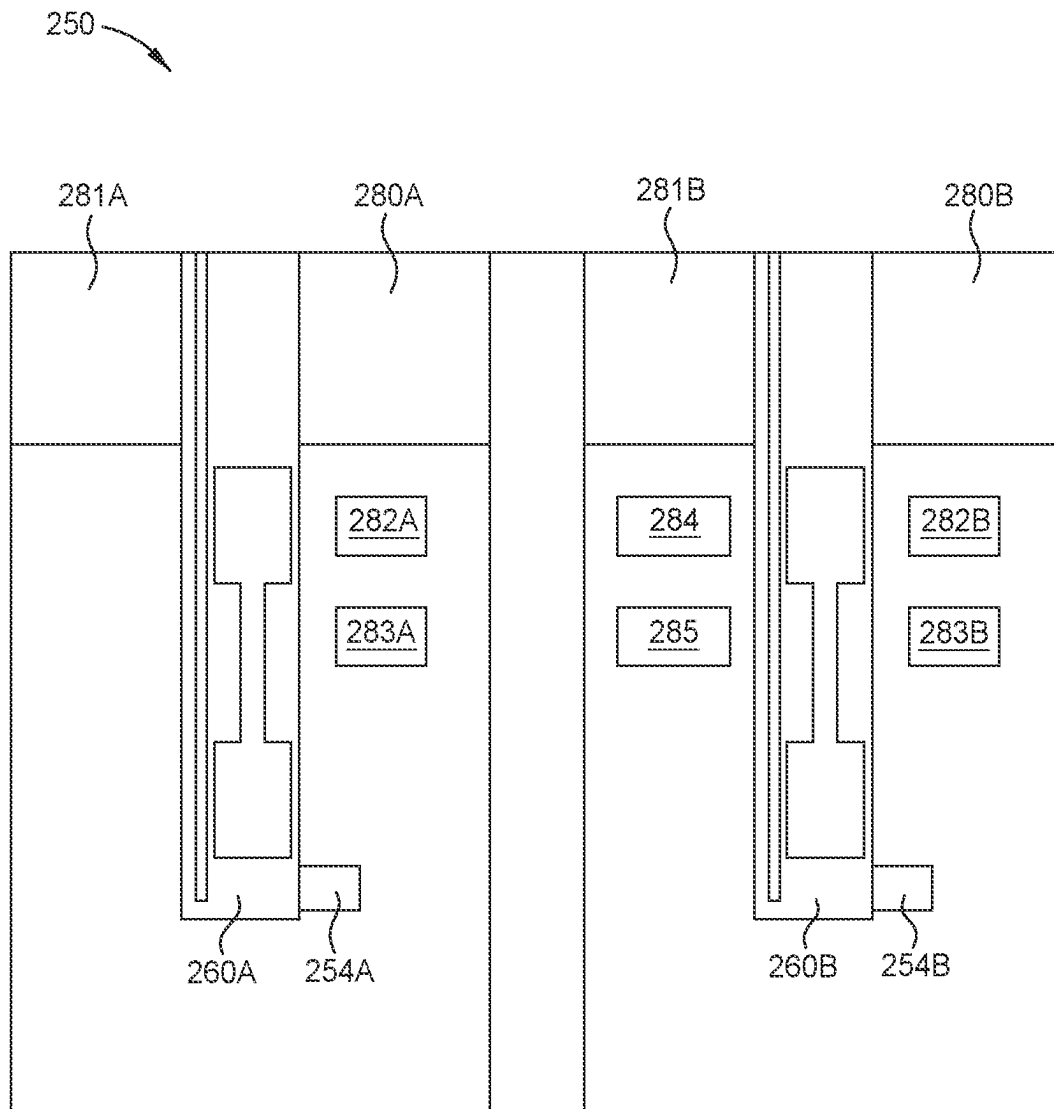


FIG. 2E

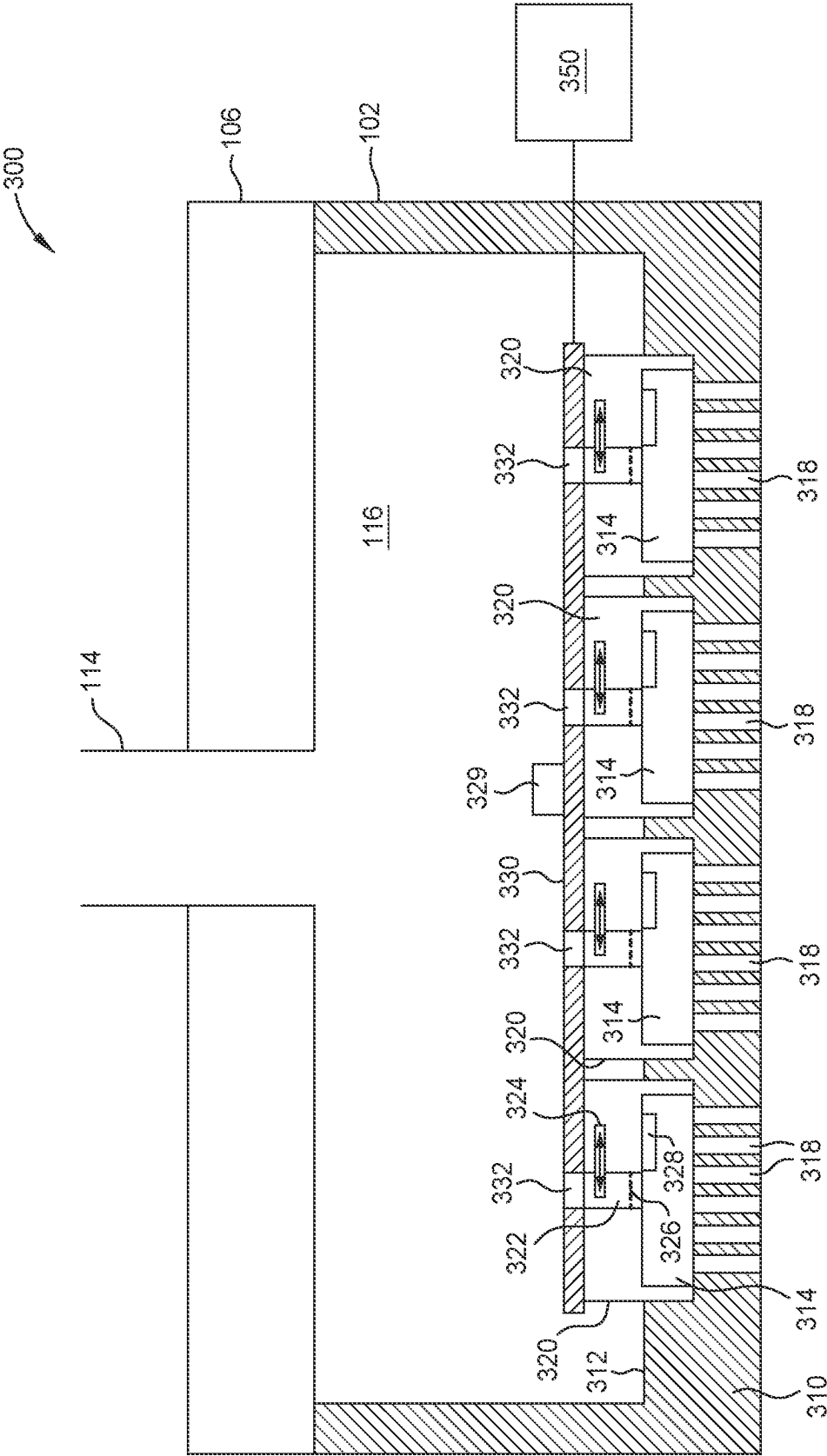


FIG. 3

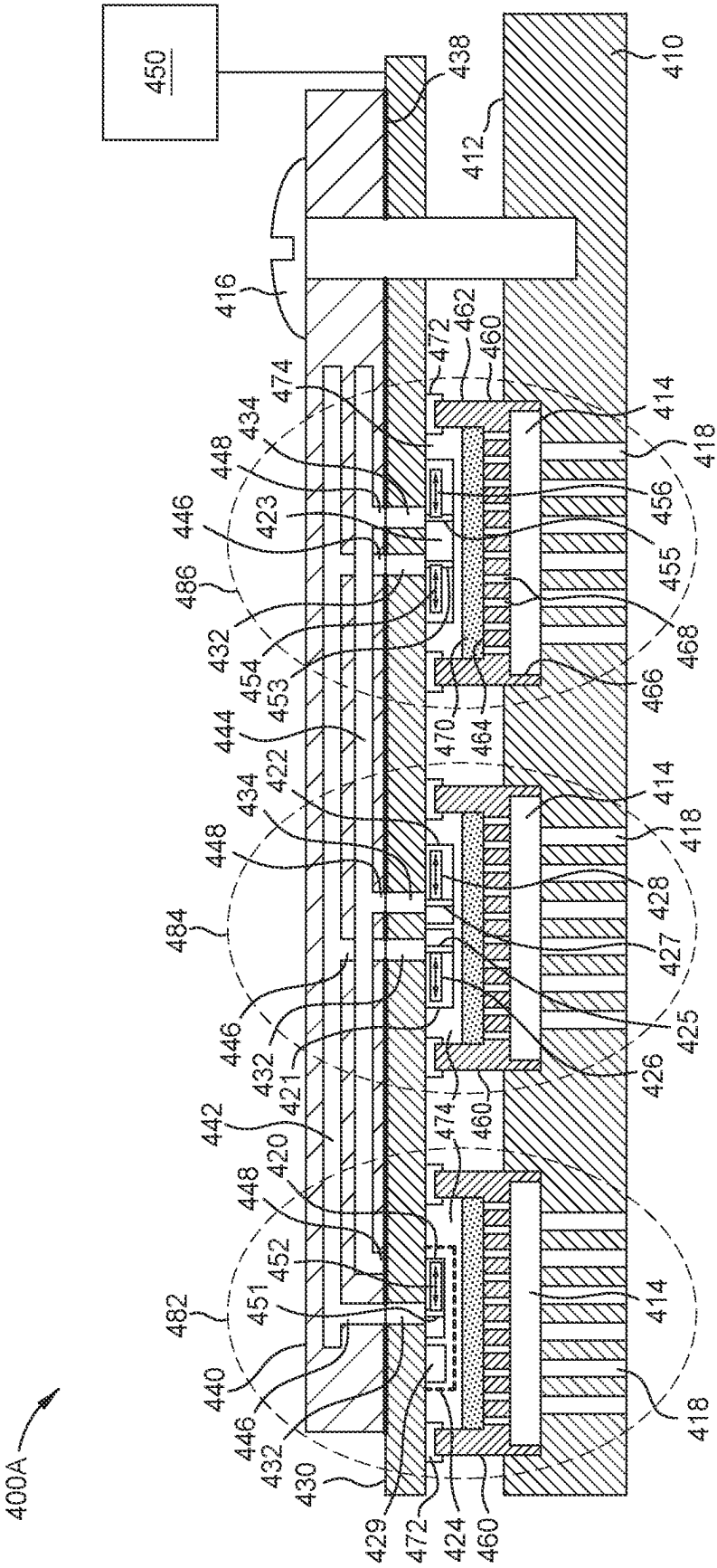
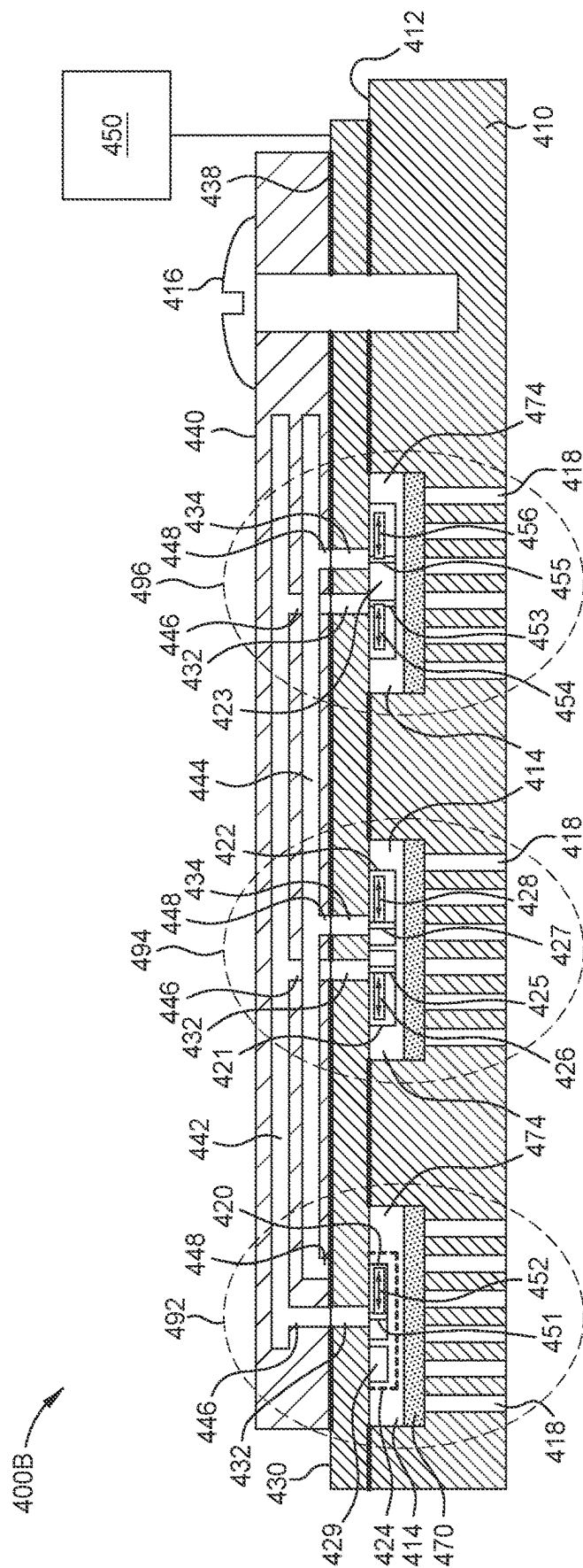


FIG. 4A



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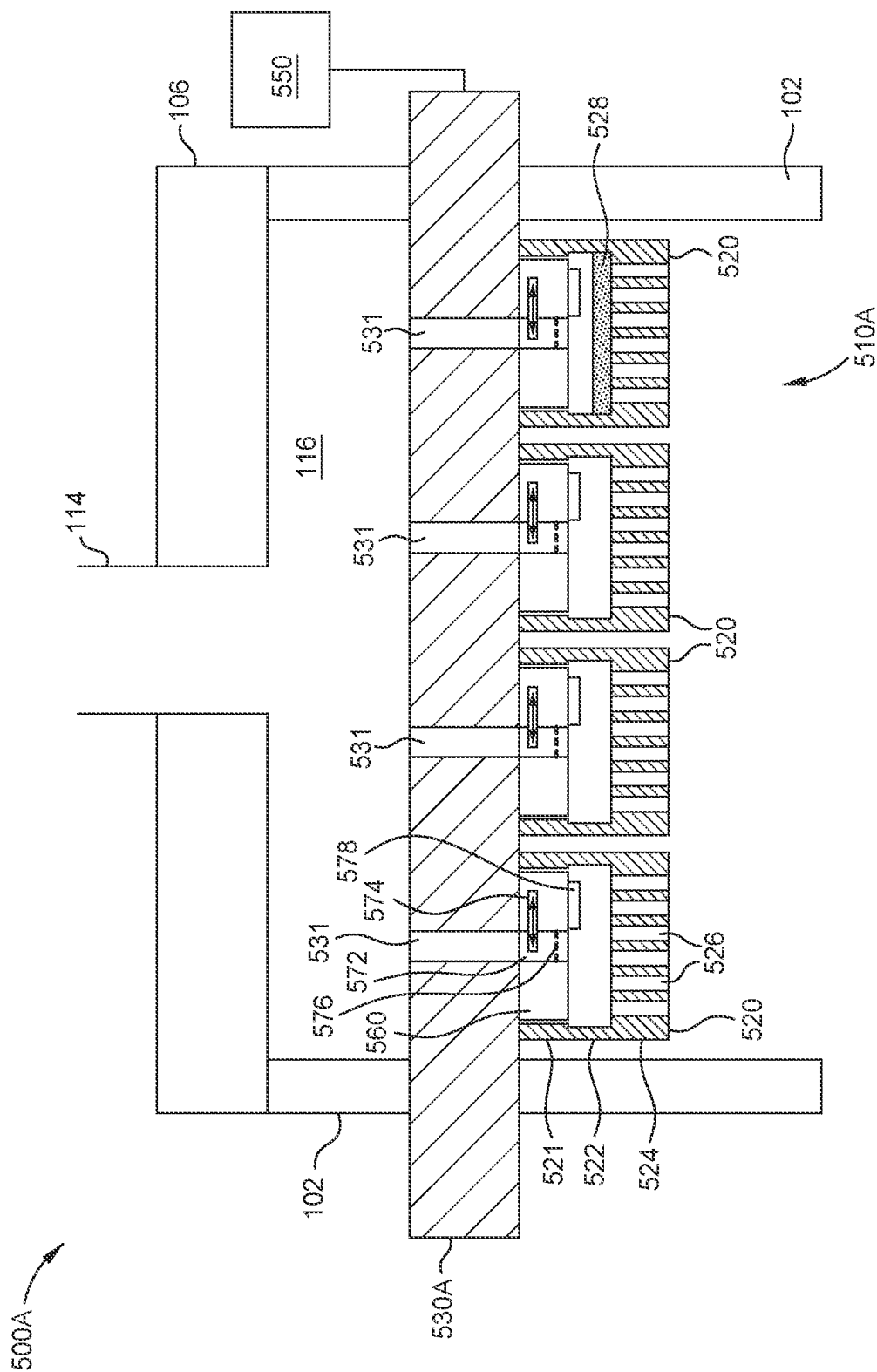


FIG. 5A

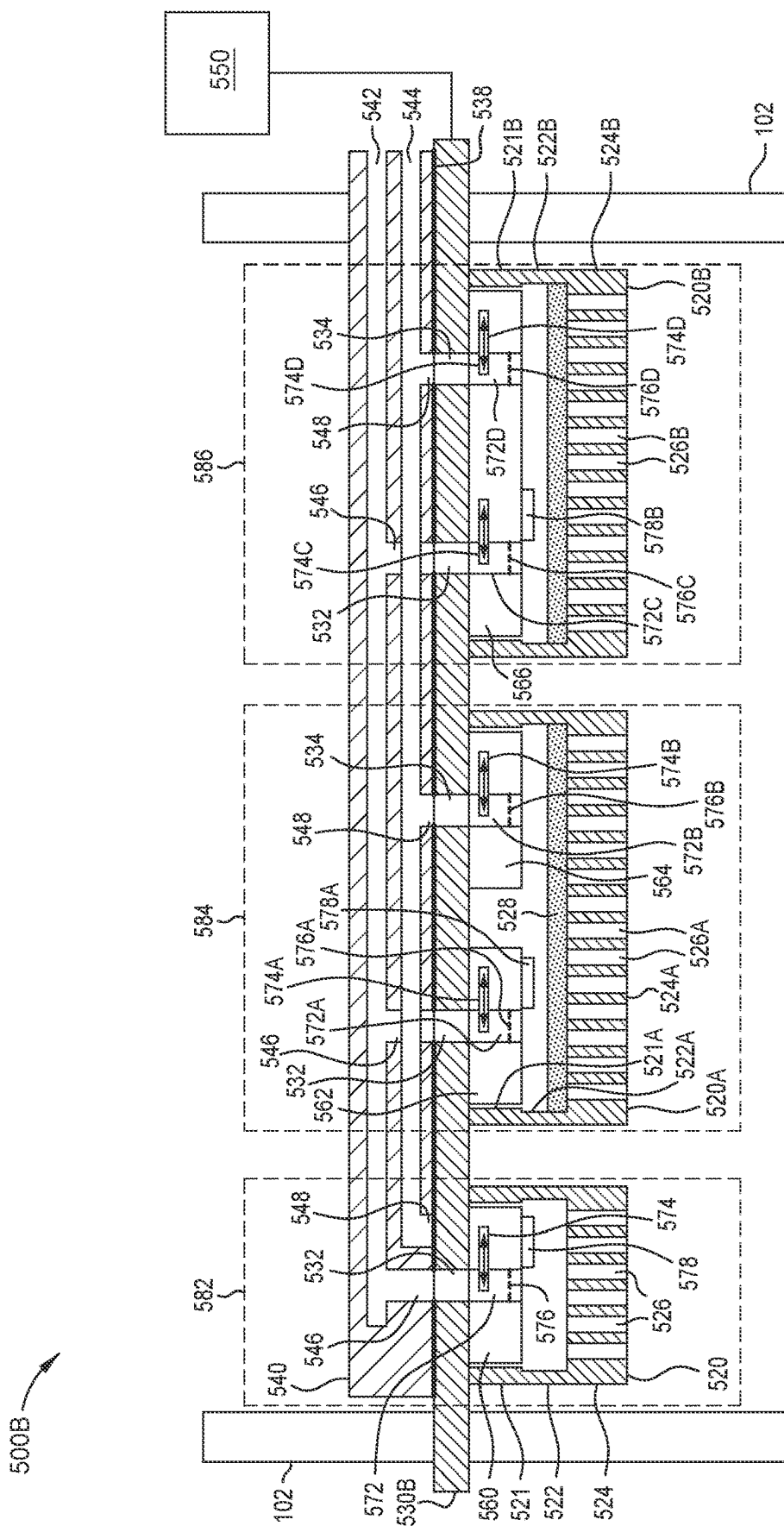


FIG. 5B

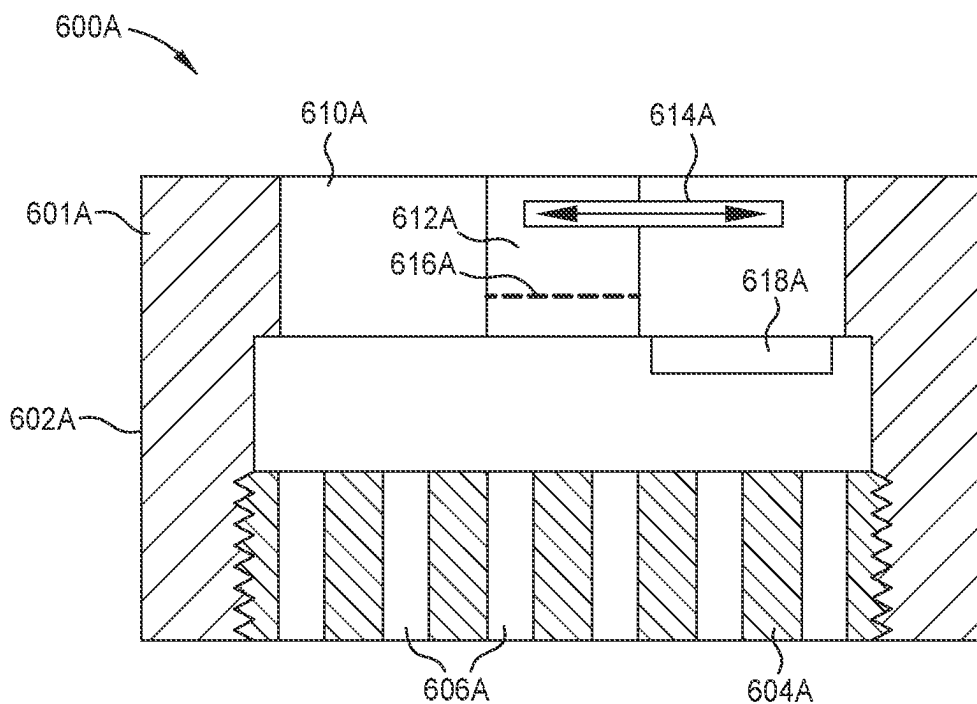


FIG. 6A

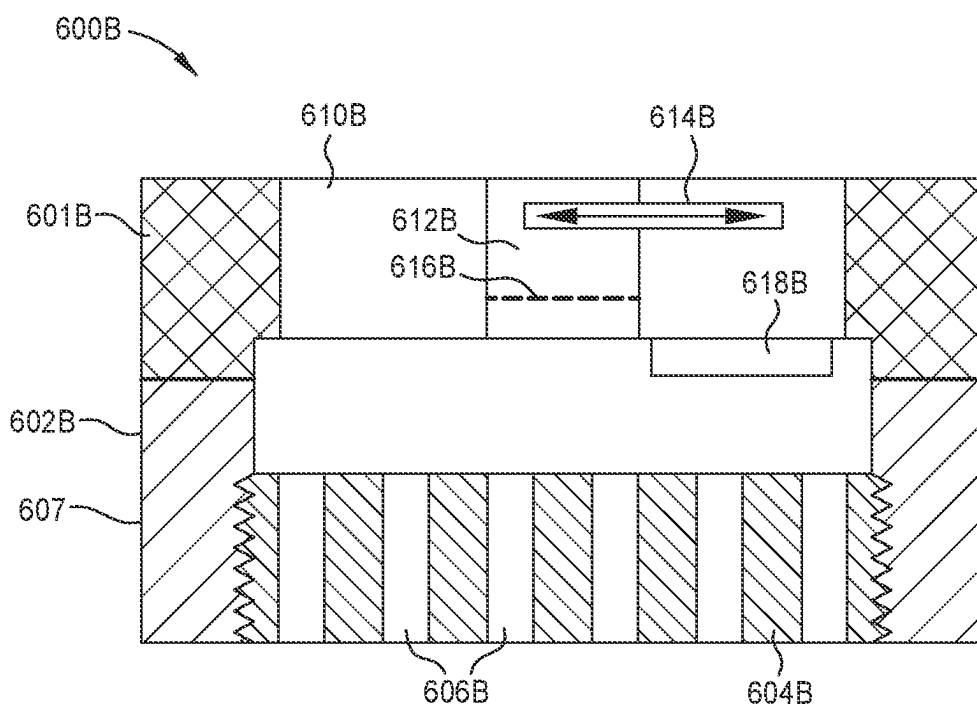


FIG. 6B

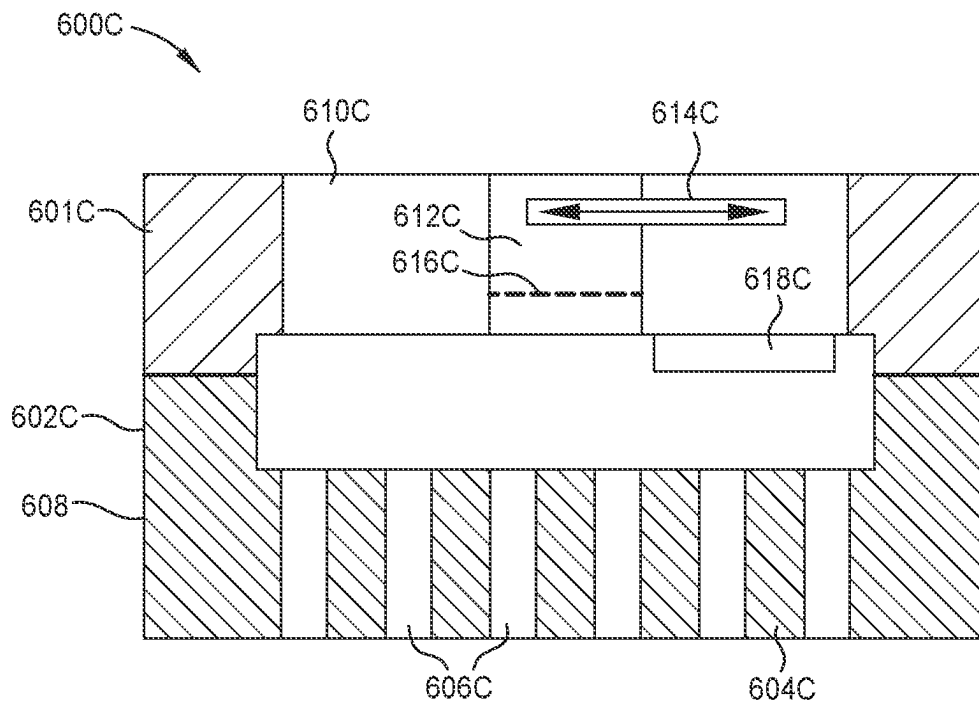


FIG. 6C

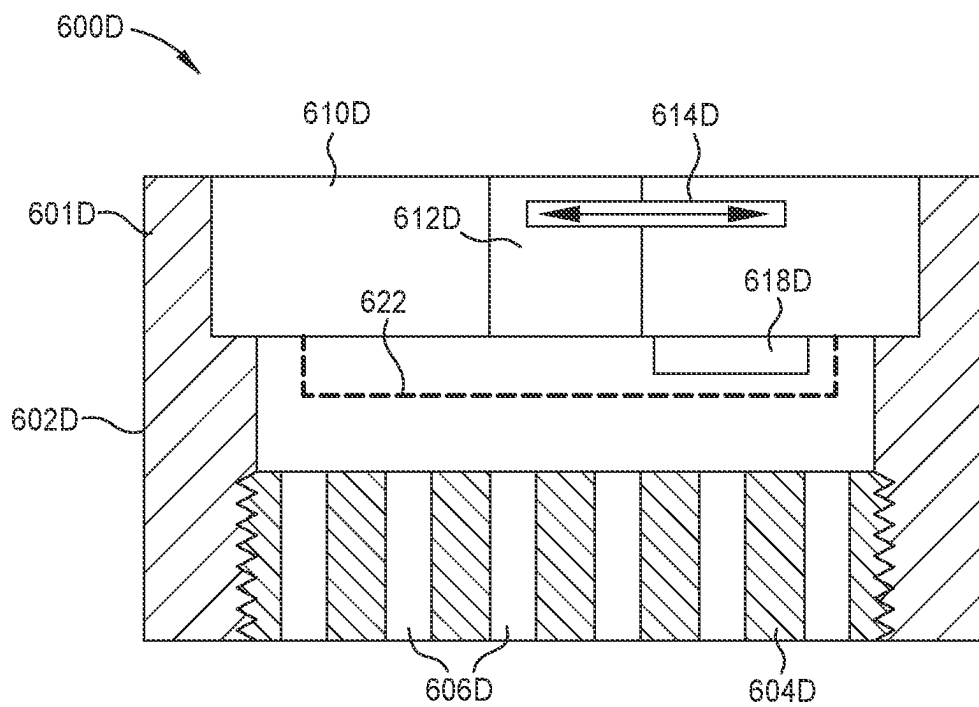


FIG. 6D

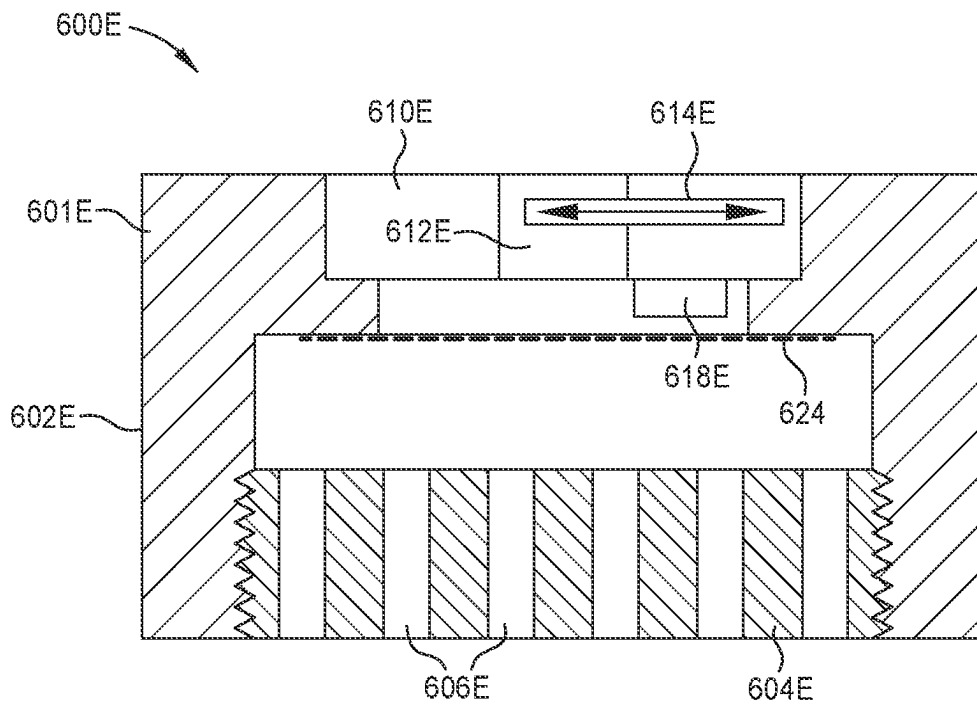


FIG. 6E

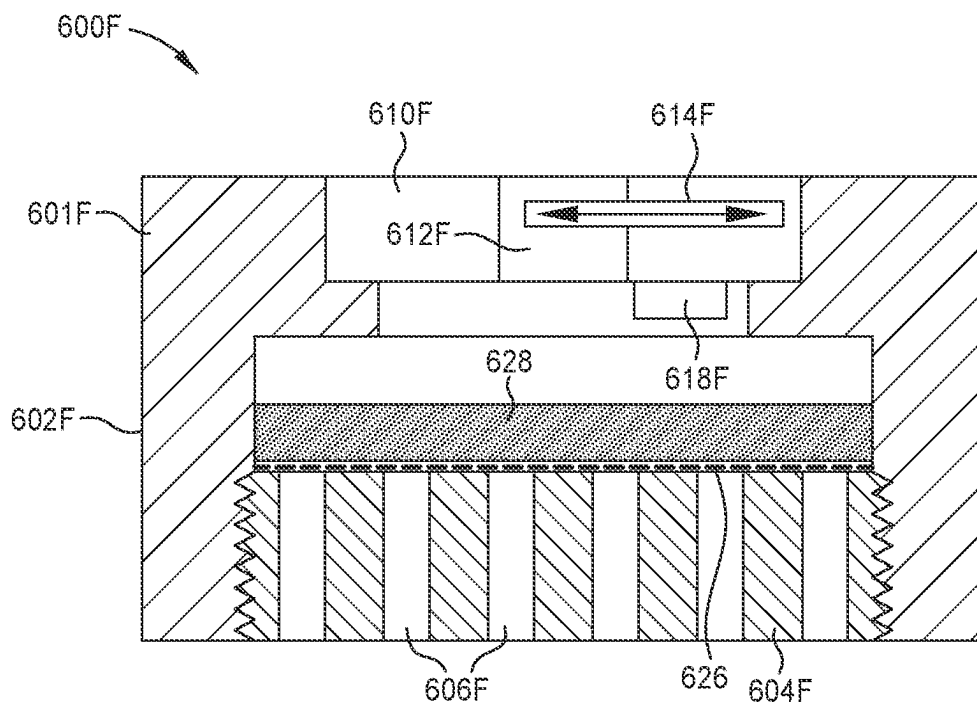


FIG. 6F

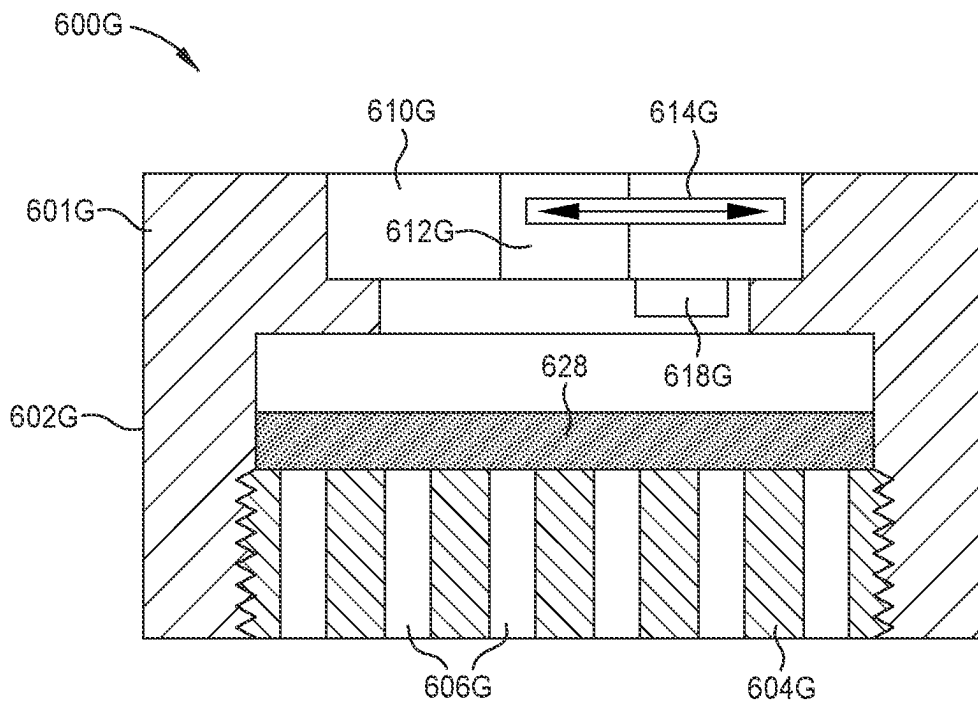


FIG. 6G

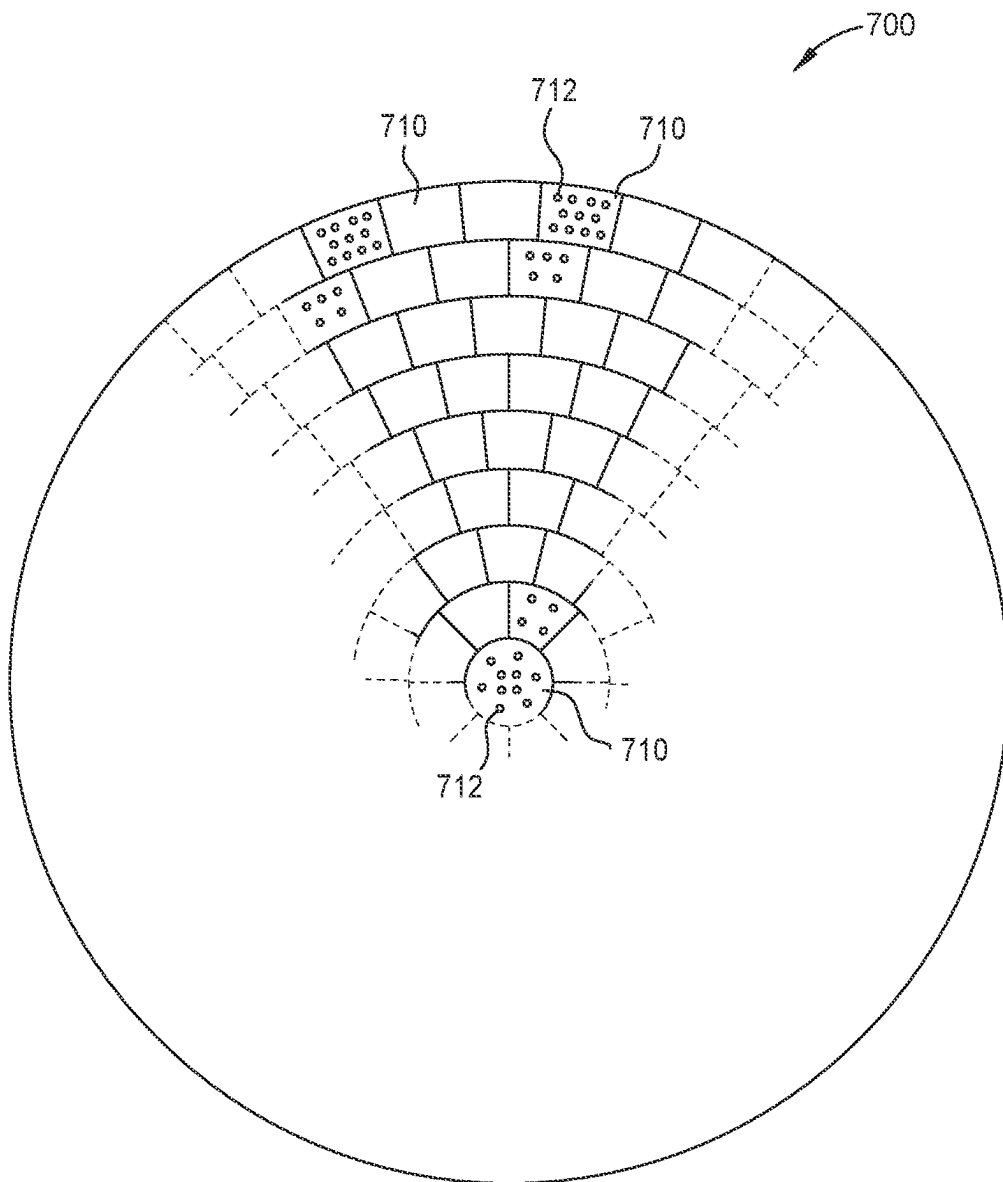


FIG. 7

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INTEGRATED SHOWERHEAD**BACKGROUND****Field**

Embodiments of the present disclosure generally relate to apparatus and methods utilized in the manufacture of semiconductor devices. More particularly, embodiments of the present disclosure relate to components, such as a showerhead, of a processing chamber for forming semiconductor devices.

Description of the Related Art

Semiconductor substrates are processed for a wide variety of applications, including the fabrication of integrated devices and microdevices. During processing, the substrate is positioned on a substrate support within a process chamber. The interior of the process chamber is placed under vacuum while the substrate is processed by exposure to process gases. Some processes involve etching material away from the substrate, and other processes involve the deposition of material onto the substrate. The uniformity of the etch or of the material deposited on the substrate may be affected by the distribution of process gases within the process chamber. In some process chambers, a showerhead distributes the process gas. The pattern and sizes of holes in the showerhead may be optimized for the distribution of a certain process gas for a particular processing operation, but may not be optimized for a different process gas or for distributing process gasses in a different processing operation.

Thus, there is a need for improved process chambers that facilitate effective control over process gas distribution.

SUMMARY

The present disclosure generally relates to components, such as a showerhead, of a substrate processing chamber for forming semiconductor devices. In one embodiment, a showerhead for a processing chamber includes a faceplate. The faceplate includes a bottom surface, a top surface, and a plurality of openings extending from the top surface to the bottom surface. A printed circuit board is coupled to the faceplate. The showerhead further includes a plurality of MEMS devices coupled to the printed circuit board, each MEMS device associated with one or more unique openings of the plurality of openings, and configured to regulate a gas flow through the corresponding one or more unique openings. The showerhead further includes a plurality of local controllers coupled to the printed circuit board, each local controller configured to control operation of a corresponding MEMS device of the plurality of MEMS devices independently of an operation of other MEMS devices of the plurality of MEMS devices.

In another embodiment, a showerhead for a processing chamber includes a printed circuit board including a plurality of ports therethrough. The showerhead further includes a faceplate. The faceplate includes a plurality of MEMS modules coupled to the printed circuit board. Each MEMS module includes: a body; sidewalls extending below the body to a base, the base including one or more holes; and a MEMS device operable to control gas flow through at least one of the plurality of ports.

In another embodiment, a processing chamber includes a chamber body and a showerhead disposed in the chamber

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body. The faceplate includes a bottom surface, a top surface, a plurality of compartments recessed into the top surface, and a plurality of openings extending from each compartment to the bottom surface. The showerhead further includes a plurality of MEMS devices, each MEMS device in a corresponding compartment of the plurality of compartments, and configured to regulate a gas flow into each corresponding compartment. The showerhead further includes a printed circuit board coupled to the top surface of the faceplate and to each MEMS device. The showerhead further includes a controller coupled to the printed circuit board, and configured to control operations of at least one MEMS device of the plurality of MEMS devices independently of operations of other MEMS devices of the plurality of MEMS devices.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, as the disclosure may admit to other equally effective embodiments.

FIG. 1 is a schematic cross-sectional view of a processing chamber.

FIG. 2A is an isometric view of the top of an exemplary MEMS device.

FIG. 2B is an isometric view of the bottom of the MEMS device of FIG. 2A.

FIGS. 2C and 2D are plan views of the top of the MEMS device of FIG. 2A.

FIG. 2E is a plan view of the top of another exemplary MEMS device.

FIG. 3 is a schematic cross-sectional side view of an exemplary showerhead.

FIGS. 4A and 4B are schematic cross-sectional side view of exemplary showerheads.

FIGS. 5A and 5B are schematic cross-sectional side views of exemplary showerheads.

FIGS. 6A-6G are schematic cross-sectional side views of configurations of MEMS modules.

FIG. 7 is a schematic plan view of a faceplate of a showerhead.

To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in one or more other embodiments without further recitation.

DETAILED DESCRIPTION

The present disclosure concerns components, such as a showerhead, of a processing chamber for forming semiconductor devices. Embodiments of the present disclosure provide showerheads that can be readily configured for use with any one or more of a plurality of gases used in the processing of substrates. Example gases include silicon-containing gases, oxygen-containing gases, nitrogen-containing gases, hydrogen-containing gases, argon-containing gases, and metal-containing gases.

FIG. 1 illustrates a schematic cross-sectional view of a processing chamber 100. As illustrated, the processing

chamber **100** is configured as a Plasma Enhanced Chemical Vapor Deposition (PECVD) chamber, although in some embodiments, processing chamber **100** may be configured to perform another plasma-enhanced processing operation (such as etching or physical vapor deposition) or a processing operation that does not involve plasma (such as chemical vapor deposition). The processing chamber **100** features a chamber body **102**, a substrate support **104** disposed inside the chamber body **102**, and a lid **106** coupled to the chamber body **102**, and enclosing the substrate support **104** in a processing volume **120**. The substrate support **104** is configured to support a substrate **154** thereon during processing. The substrate **154** is provided to the processing volume **120** through an opening **126**. While the embodiment of FIG. **1** is directed to a PECVD chamber, the lid **106** and substrate support **104** of FIG. **1** may be used with other processing chambers that utilize plasma generated in the processing volume **120**. Additionally, the lid **106** and substrate support **104** of FIG. **1** may be used with other processing chambers that do not utilize plasma generated in the processing volume **120**.

As illustrated, a showerhead **112** contains or serves as an electrode, and is coupled to a power source **128** through a match circuit. The power source **128** is a radio frequency (RF) power source that is electrically coupled to the electrode **108**. Further, the power source **128** provides between about 100 Watts and about 3,000 Watts at a frequency of about 50 kHz to about 13.6 MHz. In some embodiments, the power source **128** can be pulsed during various operations. The electrode **108** and power source **128** facilitate control of a plasma formed within the processing volume **120**.

The substrate support **104** contains, or is formed from, one or more metallic or ceramic materials. Exemplary metallic or ceramic materials include one or more metals, metal oxides, metal nitrides, metal oxynitrides, or any combination thereof. For example, the substrate support **104** may contain or be formed from aluminum, aluminum oxide, aluminum nitride, aluminum oxynitride, or any combination thereof.

As illustrated, an electrode **122** is embedded within the substrate support **104**, but alternatively may be coupled to a surface of the substrate support **104**. The electrode **122** is coupled to a power source **136**. It is contemplated that the power source **136** may be DC power, pulsed DC power, radio frequency (RF) power, pulsed RF power, or any combination thereof. The power source **136** is configured to drive the electrode **122** with a drive signal to generate a plasma within the processing volume **120**. It is contemplated that the drive signal may be one of a DC signal and a varying voltage signal (e.g., RF signal). Further, the electrode **122** may alternatively be coupled to the power source **128** instead of the power source **136**, and the power source **136** may be omitted.

Plasma is generated in the processing volume **120** via the power source **128** and the power source **136**. An RF field is created by driving at least one of the showerhead **112** electrode and the electrode **122** with drive signals to facilitate the formation of a plasma within the processing volume **120**. The presence of a plasma facilitates processing of the substrate **154**, for example, the deposition of a film onto a surface of the substrate **154** or the etching of material from a surface of the substrate **154**.

An exhaust port **156** is coupled to a vacuum pump **157**. The vacuum pump **157** removes excess process gases or by-products from the processing volume **120** via the exhaust port **156** during and/or after processing.

A gas supply source **111** includes one or more gas sources. The gas supply source **111** is configured to deliver the one or more gases from the one or more gas sources through the showerhead **112** and into the processing volume **120**. Each of the one or more gas sources provides a process gas such as silane, disilane, tetraethyl orthosilicate (TEOS), germane, a metal halide (such as titanium tetrachloride, tantalum pentachloride, tungsten hexafluoride), an organometallic (such as tetrakis(dimethylamido) titanium, pentakis(dimethylamido) tantalum), ammonia, oxygen (O₂), hydrogen peroxide, hydrogen, diborane, chlorine (Cl₂), sulfur hexafluoride, a hydrocarbon (generically C_xH_y), among others. In some embodiments, the process gas may be ionized to form a plasma within the processing volume **120**. For example, one or more of a carrier gas and an ionizable process gas are provided into the processing volume **120** to process the substrate **154**. When processing a 300 mm substrate, the process gases are introduced to the processing chamber **100** at a flow rate from about 6500 sccm to about 8000 sccm, from about 100 sccm to about 10,000 sccm, or from about 100 sccm to about 1000 sccm. Alternatively, other flow rates may be utilized. In some examples, a remote plasma source can be used to deliver plasma to the processing chamber **100** and can be coupled to the gas supply source **111**.

The showerhead **112** features openings **118** for admitting a process gas or gases into the processing volume **120** from the gas supply source **111**. The process gases are supplied to the processing chamber **100** via the gas feed **114**, and the process gases enter a plenum **116** prior to flowing through the openings **118**. In some embodiments, different process gases that are flowed simultaneously during a processing operation enter the processing chamber **100** via separate gas feeds and separate plenums prior to entering the processing volume **120** through the showerhead **112**.

Gas flow through the openings **118** of the showerhead **112** is regulated by one or more micro-electro-mechanical systems (MEMS devices) disposed in the showerhead **112**. In some embodiments, it is contemplated that gas flow through an individual opening **118** and/or through a cluster of openings **118** may be regulated by a MEMS device. In some embodiments, it is contemplated that gas flow through all openings **118** is regulated by a plurality of MEMS devices. In an example, each MEMS device regulates gas flow through one or more openings **118** such that gas flow through any single opening **118** is regulated by a corresponding MEMS device. It is contemplated that the regulation of gas flow by a MEMS device includes permitting a maximum flow of gas through an individual opening **118** and/or through a cluster of openings **118**. It is contemplated that the regulation of gas flow by a MEMS device includes preventing any flow of gas through an individual opening **118** and/or through a cluster of openings **118**. It is contemplated that the regulation of gas flow by a MEMS device includes controlling a flow of gas through an individual opening **118** and/or through a cluster of openings **118** such that the flow is greater than zero and less than a maximum flow of gas through the individual opening **118** and/or through the cluster of openings **118**.

FIGS. 2A-2D illustrate an exemplary MEMS device **200**. FIG. 2A is an isometric view of the top of the MEMS device **200**, and FIG. 2B is an isometric view of the bottom of the MEMS device **200**. The MEMS device **200** includes a body **202**, having an orifice **204** therethrough. Although illustrated as substantially rectangular, it is contemplated that the orifice **204** may have any appropriate cross-sectional shape, such as circular, elliptical, triangular, and the like. Furthermore, in some embodiments, the orifice **204** may include

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multiple orifices. Moreover, it is contemplated that one variant of MEMS device 200 may have an orifice 204 that has a cross-sectional size different from the cross-sectional size of the orifice 204 of another variant of MEMS device 200.

A skirt 206 extends from the body 202 at the bottom of the MEMS device 200. A valve member 210 is mounted on the body 202 and regulates fluid flow through the orifice 204. The valve member 210 is electrically conductive. In some embodiments, the valve member 210 is metallic. The valve member 210 passes an electrical current that flows between contacts 230, 231, to which the valve member 210 is connected. As illustrated, in some embodiments, the MEMS device 200 includes contacts 232, 233 that are configured for connection to a heater, such as a wire through which an electrical current is passed in order to induce heating. In some of such embodiments, the heater is integrated with the MEMS device 200. Alternatively, the heater may be a separate component configured to be plugged into the MEMS device 200. In some embodiments, the heater may be omitted.

As illustrated, in some embodiments, the MEMS device 200 includes contacts 234, 235 that are configured for connection to a sensor 236. In some of such embodiments, the sensor 236 is integrated with the MEMS device 200. Alternatively, the sensor 236 may be a separate component configured to be plugged into the MEMS device 200. It is contemplated that the sensor 236 may be configured to measure one or more of pressure, temperature, or flow rate. In an example, measurements of the flow rate of a fluid through the orifice 204 may be derived at least in part from measurements of pressure obtained from the sensor 236. In some embodiments, the sensor 236 may be omitted.

FIG. 2C is a plan view of the top of the MEMS device 200. The valve member 210 of the MEMS device 200 includes a first section 212 and a second section 214. At an end 216 of the valve member 210 near the orifice 204, the first and second sections 212, 214 are connected together. At an opposite end 218 of the valve member 210, the first section 212 is connected to contact 230, and the second section 214 is connected to contact 231, but the first 212 and second 214 sections are not connected together. The first section 212 is nominally thicker than the second section 214, but includes a void 220. The void 220 is illustrated as two connected rectangles, however, it is contemplated that the void 220 may have or include any suitable shape, such as one or more triangles, one or more squares, one or more circles, one or more ellipses, or one or more of any other shape. The void 220 divides the first section into a relatively thick portion 222 and one or more relatively thin portions 224. An end portion 226 of the first section 212 is relatively thick, and is positioned at the end 216 near to the orifice 204. In the configuration illustrated, the orifice 204 is at least partially uncovered by the end portion 226, thereby permitting gas to flow through the orifice 204. The end portion 226 is configured to at least partially obscure the orifice 204 during operation of the MEMS device 200.

FIG. 2D is a plan view of the top of the MEMS device 200 when an electrical current is passed through the valve member 210. Because the first section 212 and second section 214 are connected together at end 216 but not connected together at end 218, a voltage applied at the contacts 230, 231 causes a current to flow through the first section 212 and through the second section 214. The thicknesses of the first 212 and second 214 sections impact the electrical resistances of the first and second sections 212, 214—the thicker the section, the lower the resistance. When

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a current flows through the valve member 210, the second section 214 and the relatively thin portions 224 of the first section 212 experience greater heating than the relatively thick portion(s) 222 of the first section 212.

Because of the difference in heating, the relatively thick portion(s) 222 of the first section 212 do not experience as much thermal expansion as do the second section 214 and the relatively thin portions 224 of the first section 212. Thus, the first section 212 does not linearly elongate to the same extent as does the second section 214. Because the first section 212 and the second section 214 are connected together at end 216, elongation of the second section 214 causes the first section 212 to deform into an “S” shape, facilitated by the void 220. The end 216 deflects in the direction of arrow 228, thus causing the end portion 226 of the first section 212 to at least partially obscure the orifice 204. In some embodiments, it is contemplated that the end 216 deflects to such an extent that the end portion 226 completely obscures the orifice 204. In some of such embodiments, the end portion 226 completely blocks passage of gas through the orifice 204.

As illustrated in FIG. 2D, in some embodiments, elongation of the valve member 210 due to heating is compensated by deformation of the first section 212 into the “S” shape. In this way, the orientation of the end portion 226 is maintained during travel, and the end portion 226 can completely obscure the orifice 204. However, in some embodiments, the end portion 226 does not completely obscure the orifice 204, but only partially obscures the orifice 204 when an electrical current is applied to the valve member 210.

When the current passing through the valve member 210 is reduced to a smaller magnitude, or is completely ceased, the valve member 210 cools down, experiences thermal contraction, and returns towards the position illustrated in FIG. 2C. Thus, the positioning of the end portion 226 with respect to the orifice 204 is controlled by modifying the current passing through the valve member 210. Consequently, the amount of fluid flow through the orifice 204 is controlled by adjusting the magnitude of the current passing through the valve member 210. In an example, the orifice 204 is completely uncovered at zero current through the valve member 210, the orifice 204 is completely covered by the end portion 226 at a prescribed maximum current through the valve member 210, and the orifice 204 is partially covered by the end portion 226 at a given fraction of the prescribed maximum current through the valve member 210. In such an example, a maximum fluid flowrate through the orifice 204 is realized at zero current through the valve member 210, zero flow through the orifice 204 is realized at the prescribed maximum current through the valve member 210, and a fraction of the maximum fluid flowrate through the orifice 204 is realized at the given fraction of the prescribed maximum current through the valve member 210.

In some embodiments, the applied current through the valve member 210 may be adjusted in steps, thereby providing one or more intermediate positions of the end portion 226 of the valve member 210 between fully uncovering the orifice 204 and completely obscuring the orifice 204. In such embodiments, the MEMS device 200 provides an intermediate fluid flowrate between zero flow and the maximum flowrate corresponding to each intermediate position of the end portion 226. In an example, the resulting fluid flowrate through the orifice 204 may be varied in increments (such as in 5% increments, 10% increments, or 20% increments) from zero to the maximum flow.

In some embodiments, the applied current through the valve member **210** may be continuously variable, thereby providing a continuously variable position of the end portion **226** of the valve member **210** between fully uncovering the orifice **204** and completely obscuring the orifice **204**. In such embodiments, the MEMS device **200** provides a continuously variable fluid flowrate through the orifice **204** between zero flow and the maximum flowrate, the resulting flowrate corresponding to the intermediate position of the end portion **226**.

In some embodiments, the applied current through the valve member **210** may be adjusted stepwise over a portion of the range from zero to maximum current, and may be continuously variable over another portion of the range from zero to maximum current. In such embodiments, the resulting fluid flowrate through the orifice **204** may be varied in steps over a portion of the range from zero to maximum flow, and may be continuously variable over another portion of the range from zero to maximum flow. In an example, the resulting fluid flowrate through the orifice **204** may be varied in steps from zero to 20% of the maximum flow, and may be varied continuously from 20% to the maximum flow.

Although FIGS. 2A-2D illustrate the MEMS device **200** to be configured with the orifice **204** normally open with no current applied, in some embodiments, the MEMS device **200** may be configured with the orifice **204** normally closed with no current applied. In such embodiments, the starting position for the valve member **210** includes the end portion **226** obscuring the orifice **204**. In an example, the end portion **226** completely blocks passage of gas through the orifice **204**. The application of a current through the valve member **210** causes deflection of the end **216** of the valve member, moving the end portion **226** to at least partially uncover the orifice **204**, thereby permitting gas to flow through the orifice **204**.

Any of the arrangements of MEMS devices in the present disclosure may include MEMS devices configured with a normally open orifice. Any of the arrangements of MEMS devices in the present disclosure may include MEMS devices configured with a normally closed orifice. Any of the arrangements of MEMS devices in the present disclosure may include a combination of MEMS devices configured with a normally open orifice and MEMS devices configured with a normally closed orifice.

FIG. 2E is a plan view of the top of an exemplary MEMS device **250**. MEMS device **250** is similar to MEMS device **200**, but includes two orifices **254A**, **254B** and two valve members **260A**, **260B**. Valve member **260A** is connected to contacts **280A**, **281A**, and regulates fluid flow through orifice **254A**. Valve member **260B** is connected to contacts **280B**, **281B**, and regulates fluid flow through orifice **254B**. In some embodiments, a dedicated heater is associated with each orifice **254A**, **254B**; the heater associated with orifice **254A** is connected to contacts **282A**, **283A**, and the heater associated with orifice **254B** is connected to contacts **282B**, **283B**. A sensor (**236**, FIG. 2B), as described above, is connected to contacts **284**, **285**. In some embodiments, it is contemplated that valve member **260A** and valve member **260B** may be operated independently, and therefore orifice **254A** and orifice **254B** may be suitable for coupling to separate gas supplies.

To inhibit corrosion and/or reduce a probability that a valve member of a MEMS device may stick in position and become inoperable, it is contemplated that surfaces of each component of MEMS devices **200**, **250** may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-

stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as Al_2O_3), yttrium oxides (such as Y_2O_3), silicon oxides (such as SiO_x), titanium oxides (such as TiO_2), and the like.

FIG. 3 is a schematic cross-sectional side view of an exemplary showerhead **300**. It is contemplated that the configuration of showerhead **300** may be used as showerhead **112** of FIG. 1. Showerhead **300** includes a faceplate **310** with openings **318**, through which gases flow from the plenum **116** into the processing volume (**120**, FIG. 1) of a processing chamber, such as processing chamber **100**. A top surface **312** of the faceplate **310** includes compartments **314**. As illustrated, each compartment **314** is recessed into the top surface **312**. In some embodiments, the compartments **314** may not be recessed into the top surface **312**. A cluster of openings **318** is associated with each compartment **314**.

A MEMS device **320** is associated with each compartment **314**. As illustrated, in embodiments in which a compartment **314** is recessed into the top surface **312** of the faceplate **310**, the MEMS device **320** may be at least partially disposed in a corresponding compartment **314**. It is contemplated that the MEMS device **320** may be configured similarly to MEMS device **200** or MEMS device **250**. Each MEMS device **320** is schematically depicted to include an orifice **322**, a valve member **324**, a heater **326**, and a sensor **328**, such as described above for MEMS device **200**. Each MEMS device **320** is coupled to a printed circuit board (PCB) **330**. In some embodiments, each MEMS device **320** is soldered to the PCB **330**. In some of such embodiments, the solder surrounds the orifice **322** and provides a seal between the PCB **330** and each MEMS device **320**. Each contact of each MEMS device **320** is connected to the PCB **330**. The sensor **328**, the heater **326**, and the valve member **324** of each MEMS device **320** receive electrical power via the PCB **330**. The PCB **330** is coupled to a master controller **350** for the transmission of power and/or control signals and/or telemetry with each MEMS device **320**.

The PCB **330** includes a port **332** associated with each MEMS device **320**. When the valve member **324** of a MEMS device **320** permits gas to flow through the corresponding orifice **322**, gas in the plenum **116** can flow through the corresponding port **332** in the PCB **330** and through the orifice **322** into the corresponding compartment **314** of the faceplate **310**. In some embodiments, the gas is heated by the heater **326**. The gas flows from the compartment **314** through the corresponding openings **318** in the faceplate **310** into the processing volume (**120**, FIG. 1) of the processing chamber.

As illustrated, in some embodiments, showerhead **300** may include one or more local controllers **329**. Each local controller **329** may be associated with, and programmed to control, a corresponding single MEMS device **320** or a corresponding group of MEMS devices **320**. In an example, each local controller **329** includes an application-specific integrated circuit (ASIC). In some embodiments, each local controller **329** may be integrated into a MEMS device **320**. As illustrated, in some embodiments, each local controller **329** may be coupled to the PCB **330** separate from the MEMS device **320**. In some embodiments, the local controller **329** includes an electromagnetic shield. To inhibit corrosion, it is contemplated that surfaces of the local controller **329** may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides

(such as Al_2O_3), yttrium oxides (such as Y_2O_3), silicon oxides (such as SiO_2), titanium oxides (such as TiO_2), and the like.

In some embodiments, the local controller 329 receives commands from master controller 350 via the PCB 330. It is contemplated that the commands may be in the form of a signal that is addressed to correspond with a specific device, such as a specific MEMS device 320. Each local controller 329 is programmed to recognize command signals addressed to correspond with devices under the purview of local controller 329, and controls the devices according to the commands received. In some embodiments, each local controller 329 is programmed to ignore command signals that are not addressed to correspond with any of the devices under the purview of local controller 329.

In some embodiments, each MEMS device 320 is independently addressable via a corresponding local controller 329, such that the operation of each MEMS device 320 can be controlled without changing the operating status of any other MEMS device 320. In some embodiments, each MEMS device 320 is assigned to one or more groups of MEMS devices 320, and each group of MEMS devices 320 is independently addressable via one or more corresponding local controllers 329. In such embodiments, the operation of each MEMS device 320 within a defined group can be controlled without changing the operating status of any other MEMS device 320 that is not within the defined group.

In an example, each MEMS device 320 or group of MEMS devices 320 are associated with a discrete zone of the faceplate 310, such as illustrated by any zone 710 of faceplate 700 in FIG. 7. The control of each MEMS device 320, or group of MEMS devices 320, independently of other MEMS devices of showerhead 300 facilitates the adjustment of gas flow distribution across the zones of the faceplate 310.

In an example, a cluster of MEMS devices 320 at the center of the faceplate 310 are assigned to "Group A" and a cluster of MEMS devices 320 at an edge of the faceplate 310 are assigned to "Group B." The MEMS devices 320 of Group A can be controlled independently from the MEMS devices of Group B. Additionally, the MEMS devices 320 of Group A can be controlled via a command addressed to the group, and the MEMS devices 320 of Group B do not respond to the command addressed to Group A. In such an example, the MEMS devices 320 of Groups A and B can be controlled to adjust the quantity of process gas being delivered to the center of a substrate, such as substrate 154, relative to the quantity of process gas being delivered to the edge of the substrate.

In another example, a processing chamber, such as processing chamber 100, has an exhaust port (156, FIG. 1) located in an off-center position, which causes variations in gas flow at different locations within the processing volume (120, FIG. 1) of the processing chamber. Such azimuthal variations in gas flow can result in uneven processing of a substrate, such as a disparity of film thickness across the substrate. In such an example, a cluster of MEMS devices 320 near the exhaust port are assigned to "Group C" and a cluster of MEMS devices 320 further away from the exhaust port are assigned to "Group D." The MEMS devices 320 of Group C can be controlled independently from the MEMS devices of Group D. Additionally, the MEMS devices 320 of Group C can be controlled via a command addressed to the group, and the MEMS devices 320 of Group D do not respond to the command addressed to Group C. In such an example, the MEMS devices 320 of Groups C and D can be controlled to adjust the quantity of process gas being delivered to a portion of the substrate near to the exhaust port

relative to the quantity of process gas being delivered to a portion of the substrate further away from the exhaust port.

In some embodiments, one or more MEMS devices 320 may be controlled according to a hierarchy of commands such that MEMS devices 320 not within a specific hierarchical set of MEMS devices 320 are unaffected by operating commands addressed to MEMS devices 320 within the specific hierarchical set. In an example, a particular MEMS device 320 is allocated to a small group of MEMS devices ("Group E1") which is part of a larger group of MEMS devices ("Group E"). In this example, the particular MEMS device 320 is also allocated to a different group of MEMS devices ("Group F") that contains other MEMS devices that are not within Group E. The particular MEMS device 320 can be controlled by commands addressed only to that particular MEMS device 320, and no other MEMS device will respond to those commands. The particular MEMS device 320 can be controlled also by commands addressed only to Group E1. All MEMS devices in Group E1, including that particular MEMS device 320, will respond to those commands, but no other MEMS device will respond to those commands. The particular MEMS device 320 can be controlled also by commands addressed only to Group E. All MEMS devices in Group E, including that particular MEMS device 320, will respond to those commands, but no other MEMS device will respond to those commands. The particular MEMS device 320 can be controlled also by commands addressed only to Group F. All MEMS devices in Group F, including that particular MEMS device 320, will respond to those commands, but no other MEMS device—including MEMS devices within Group E1 or Group E, unless those other MEMS devices are also allocated to Group F—will respond to those commands.

In embodiments in which local controller 329 is omitted, master controller 350 operates each MEMS device 320 via electrically conductive lines embedded in the PCB 330.

FIG. 4A is a schematic cross-sectional side view of an exemplary showerhead 400A. It is contemplated that the configuration of showerhead 400A may be used as showerhead 112 of FIG. 1. Showerhead 400A includes a manifold 440 disposed on a PCB 430. The manifold 440 includes a first conduit 442 for passage of a first gas, and a second conduit 444 for passage of a second gas. It is contemplated that the first conduit 442 is isolated from the second conduit 444 such that the first gas and the second gas do not mix in the manifold 440. First ducts 446 from the first conduit 442 are aligned with first ports 432 in the PCB 430. Second ducts 448 from the second conduit 444 are aligned with second ports 434 in the PCB 430. An interface 438 between the manifold 440 and the PCB 430 is sealed, such as by bonding the manifold 440 to the PCB 430, to inhibit mixing of the first gas and the second gas at the interface 438.

In some embodiments, manifold 440 includes one or more additional conduits and corresponding ducts configured to convey one or more additional gases. In such embodiments, the one or more additional conduits may be isolated from the first conduit 442 and the second conduit 444. Furthermore, it is contemplated that the PCB 430 may include additional ports aligned with the additional ducts.

The showerhead 400A includes a faceplate 410 with openings 418, through which gases flow into the processing volume (120, FIG. 1) of a processing chamber, such as processing chamber 100. As illustrated, in some embodiments, the manifold 440 and PCB 430 are coupled to the faceplate 410 by one or more fastener 416, such as a screw or a bolt. A top surface 412 of the faceplate 410 includes compartments 414. As illustrated, each compartment 414 is

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recessed into the top surface **412**. In some embodiments, the compartments **414** may not be recessed into the top surface **412**. A cluster of openings **418** is associated with each compartment **414**.

A spacer **460** is associated with each compartment **414**. As illustrated, in embodiments in which a compartment **414** is recessed into the top surface **412** of the faceplate **410**, the spacer **460** may be at least partially disposed in a corresponding compartment **414**. Each spacer **460** includes sidewalls **462** and a floor **464**. Although the sidewalls **462** are illustrated as extending to form a shroud **466** below the floor **464**, in some embodiments, the shroud **466** may be omitted. Holes **468** in the floor **464** facilitate communication of gas to the openings **418** of the faceplate **410**. As illustrated, in some embodiments, a diffuser **470** is disposed above the holes **468**. The diffuser **470** can promote a uniform distribution of gas through the holes **468**. In some embodiments, the diffuser **470** filters out particles entrained in the gas. Example diffusers **470** include a mesh (such as a sintered mesh), a porous metal filter, a foam (such as porous PTFE foam), or the like. In some embodiments, the diffuser **470** may be omitted.

The sidewalls **462** of the spacer **460** extend above the floor **464** to the PCB **430**. As illustrated, in some embodiments, a gasket **472** seals an interface between the spacer **460** and the PCB **430**. The gasket **472** may be made of any suitable material that can form a pressure seal and is resistant to chemical attack, such as an elastomer/thermoplastic material (such as an FKM type material, such as polyvinylidene difluoride (PVDF), including PVDF in the form of a closed cell foam), or the like. Each spacer **460** encloses a void space **474** between the PCB **430** and each corresponding compartment **414**. The ports **432**, **434** of the PCB **430** convey gas into the void spaces **474** enclosed by each spacer **460**.

To inhibit corrosion, it is contemplated that the spacer **460** may be manufactured out of a corrosion-resistant material, such as a ceramic or a metal such as titanium. Additionally, or alternatively, surfaces of the spacer **460** may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as Al_2O_3), yttrium oxides (such as Y_2O_3), silicon oxides (such as SiO_x), titanium oxides (such as TiO_2), and the like.

It is contemplated that clusters of MEMS devices of showerhead **400A** may be controlled in similar ways to the examples provided above with respect to clusters of MEMS devices **320** of showerhead **300**.

FIG. **4A** illustrates three exemplary configurations in which one or more MEMS devices **420**, **421**, **422**, **423** with spacers **460**, and attached to the PCB **430**, can be associated with each compartment **414** of the faceplate **410**. Each pairing of one or more MEMS devices **420**, **421**, **422**, **423** with a corresponding compartment **414** may be considered as a discrete unit; each unit is described below.

In a first unit **482**, the PCB **430** includes a port **432** aligned with a first duct **446** of the manifold **440**, but does not include a port corresponding to a second duct **448** of the manifold **440**. Consequently, the first unit **482** is configured to manage gas supplied via the first conduit **442** of the manifold **440**, but is not configured to manage gas supplied via the second conduit **444** of the manifold **440**. However, in alternative embodiments, the first unit **482** may be configured to manage gas supplied via the second conduit **444** of the manifold **440**, but not configured to manage gas supplied via the first conduit **442** of the manifold **440**. In such embodiments, the PCB **430** includes a port aligned with

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a second duct **448** of the manifold **440**, but does not include a port corresponding to a first duct **446** of the manifold **440**.

A MEMS device **420** within the void space **474** is coupled to the PCB **430**, and regulates gas flow through the port **432** of the PCB **430**. It is contemplated that the MEMS device **420** may be configured similarly to MEMS device **200**. MEMS device **420** includes an orifice **451** and a valve member **452**. In some embodiments, the MEMS device **420** is soldered to the PCB **430**. In some of such embodiments, the solder surrounds the orifice **451** and provides a seal between the PCB **430** and the MEMS device **420**. In some embodiments, the MEMS device **420** includes a sensor, such as sensor **236** (FIG. **2B**). In some embodiments, a sensor separate from the MEMS device **420** is coupled to the PCB **430** within the void space **474**. Such a sensor may measure any one or more of pressure, temperature, or flow rate. In some embodiments, the MEMS device **420** includes a heater. In some embodiments, as illustrated, a heater **424** separate from the MEMS device **420** is coupled to the PCB **430** within the void space **474**. The sensor, the heater, and the valve member associated with each MEMS device **420**, and heater **424** (if present) receive electrical power via the PCB **430**.

As illustrated in the first unit **482**, in some embodiments, a local controller **429** may be associated with, and programmed to control, the MEMS device **420**. In an example, the local controller **429** includes an application-specific integrated circuit (ASIC). In some embodiments, the local controller **429** is integrated into the MEMS device **420**. In some embodiments, the local controller **429** is coupled to the PCB **430** separate from the MEMS device **420**. In some embodiments, the local controller **429** includes an electromagnetic shield. To inhibit corrosion, it is contemplated that surfaces of the local controller **429** may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as Al_2O_3), yttrium oxides (such as Y_2O_3), silicon oxides (such as SiO_x), titanium oxides (such as TiO_2), and the like.

In embodiments that include the local controller **429**, the local controller **429** controls operation of the MEMS device **420** and/or the heater **424**. In some embodiments, the local controller **429** receives commands from master controller **450** via the PCB **430**. It is contemplated that the commands may be in the form of a signal that is addressed to correspond with a specific device, such as MEMS device **420** or heater **424**. The local controller **429** in the first unit **482** is programmed to recognize command signals addressed to correspond with devices in the first unit **482** (such as MEMS device **420** or heater **424**), and controls the devices according to the commands received. In some embodiments, the local controller **429** is programmed to ignore command signals that are not addressed to correspond with any of the devices under the purview of local controller **429**, such as devices in the first unit **482**. In some embodiments, the local controller **429** may be programmed to control one or more devices that are not in the first unit **482**. In an example, the local controller **429** is programmed to control one or more MEMS devices in one or more units in addition to control MEMS device **420** and heater **424** of the first unit **482**.

In some embodiments, each device in the first unit **482** is independently addressable via a corresponding local controller **429**, such that the operation of each device in the first unit **482** can be controlled without changing the operating status of any other device of showerhead **400A**. In some embodiments, each device in the first unit **482** is assigned to

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one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers 429. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead 400A that is not within the defined group.

In embodiments in which the local controller 429 is omitted, master controller 450 operates the MEMS device 420 and the heater 424 via electrically conductive lines embedded in the PCB 430.

In a second unit 484, the PCB 430 includes a first port 432 aligned with a first duct 446 of the manifold 440, and a second port 434 aligned with a second duct 448 of the manifold 440. Consequently, the second unit 484 is configured to manage gas supplied via the first conduit 442 of the manifold 440 and via the second conduit 444 of the manifold 440.

A first MEMS device 421 in the void space 474 of the second unit 484 is coupled to the PCB 430, and regulates gas flow through the first port 432 of the PCB 430. A second MEMS device 422 in the void space 474 of the second unit 484 is coupled to the PCB 430, and regulates gas flow through the second port 434 of the PCB 430. It is contemplated that each MEMS device 421, 422 may be configured similarly to MEMS device 200. MEMS device 421 includes an orifice 425 and a valve member 426. MEMS device 422 includes an orifice 427 and a valve member 428. In some embodiments, each MEMS device 421, 422 is soldered to the PCB 430. In some of such embodiments, the solder surrounds each orifice 425, 427 and provides a seal between the PCB 430 and each MEMS device 421, 422. In some embodiments, at least one of the MEMS devices 421, 422 includes a sensor, such as sensor 236 (FIG. 2B). In some embodiments, a sensor separate from the MEMS devices 421, 422 is coupled to the PCB 430. Such a sensor may measure any one or more of pressure, temperature, or flow rate. In some embodiments, each MEMS device 421, 422 includes a heater. In some embodiments, a heater (such as heater 424 of the first unit 482) separate from the MEMS devices 421, 422 is coupled to the PCB 430.

In some embodiments, the second unit 484 includes a local controller, such as local controller 429 of the first unit 482. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS devices 421, 422 and/or a separate heater, if present) in the second unit 484. In an example, the local controller is integrated into one of the MEMS devices 421, 422. In another example, the local controller is coupled to the PCB 430 separate from the MEMS devices 421, 422.

In some embodiments, each device in the second unit 484 is independently addressable via a corresponding local controller 429, such that the operation of each device in the second unit 484 can be controlled without changing the operating status of any other device of showerhead 400A. In some embodiments, each device in the second unit 484 is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers 429. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead 400A that is not within the defined group.

In a third unit 486, the PCB 430 includes a first port 432 aligned with a first duct 446 of the manifold 440, and a second port 434 aligned with a second duct 448 of the manifold 440. Consequently, the third unit 486 is configured

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to manage gas supplied via the first conduit 442 of the manifold 440 and via the second conduit 444 of the manifold 440.

A MEMS device 423 in the void space 474 of the third unit 486 is coupled to the PCB 430, and regulates gas flow through the first port 432 of the PCB 430 and through the second port 434 of the PCB 430. It is contemplated that the MEMS device 423 may be configured similarly to MEMS device 250. MEMS device 423 includes a first orifice 453 and a first valve member 454 for controlling gas flow through the first port 432 of the PCB 430, and includes a second orifice 455 and a second valve member 456 for controlling gas flow through the second port 434 of the PCB 430. In some embodiments, the MEMS device 423 is soldered to the PCB 430. In some of such embodiments, the solder surrounds the first orifice 453 and/or the second orifice 455, and provides a seal between the PCB 430 and the MEMS device 423. In some embodiments, the MEMS device 423 includes a sensor, such as sensor 236 (FIG. 2B). In some embodiments, a sensor separate from the MEMS device 423 is coupled to the PCB 430. Such a sensor may measure any one or more of pressure, temperature, or flow rate. In some embodiments, the MEMS device 423 includes a heater associated with the first orifice 453. In some embodiments, the MEMS device 423 includes a heater associated with the second orifice 455. In some embodiments, a heater (such as heater 424 of the first unit) separate from the MEMS device 423 is coupled to the PCB 430.

In some embodiments, the third unit 486 includes a local controller, such as local controller 429 of the first unit 482. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS device 423 and/or a separate heater, if present) in the third unit 486. In an example, the local controller is integrated into the MEMS device 423. In another example, the local controller is coupled to the PCB 430 separate from the MEMS device 423.

In some embodiments, each device in the third unit 486 is independently addressable via a corresponding local controller 429, such that the operation of each device in the third unit 486 can be controlled without changing the operating status of any other device of showerhead 400A. In some embodiments, each device in the third unit 486 is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers 429. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead 400A that is not within the defined group.

FIG. 4B is a schematic cross-sectional side view of an exemplary showerhead 400B, which is a variant of showerhead 400A. It is contemplated that the configuration of showerhead 400B may be used as showerhead 112 of FIG. 1.

FIG. 4B depicts the same components as in FIG. 4A, and the above description for FIG. 4A applies for FIG. 4B except that the spacers 460 are omitted, the PCB 430 is attached directly to the top surface 412 of the faceplate 410, and the void spaces 474 are within the faceplate 410. An interface between the faceplate 410 and the PCB 430 is sealed, such as by bonding the faceplate 410 to the PCB 430. As illustrated, in some embodiments, the manifold 440 and PCB 430 are coupled to the faceplate 410 by one or more fastener 416, such as a screw or a bolt. In some embodiments, the fastener 416 may be omitted.

First unit 492, second unit 494, and third unit 496 correspond to the first unit 482, second unit 484, and third unit

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486, respectively, of FIG. 4A. As illustrated, in some embodiments, a diffuser 470 is disposed in each compartment 414 above the openings 418 in the faceplate 410. In some embodiments, the diffuser 470 may be omitted.

It is contemplated that clusters of MEMS devices of showerhead 400B may be controlled in similar ways to the examples provided above with respect to clusters of MEMS devices 320 of showerhead 300.

As illustrated, in some embodiments, the first unit 492 includes local controller 429, such as described above. In such embodiments, it is contemplated that the local controller 429 controls operation of at least one device (such as the MEMS device 420 and/or a separate heater 424, if present) in the first unit 492. In an example, the local controller 429 is integrated into the MEMS device 420. In some embodiments, the local controller 429 is coupled to the PCB 430 separate from the MEMS device 420. To inhibit corrosion, it is contemplated that surfaces of the local controller 429 may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as Al_2O_3), yttrium oxides (such as Y_2O_3), silicon oxides (such as SiO_x), titanium oxides (such as TiO_2), and the like.

In some embodiments, each device in the first unit 492 is independently addressable via a corresponding local controller 429, such that the operation of each device in the first unit 492 can be controlled without changing the operating status of any other device of showerhead 400B. In some embodiments, each device in the first unit 492 is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers 429. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead 400B that is not within the defined group.

In some embodiments, the second unit 494 includes a local controller, such as local controller 429 of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS devices 421, 422 and/or a separate heater, if present) in the second unit 494. In an example, the local controller is integrated into one of the MEMS devices 421, 422. In another example, the local controller is coupled to the PCB 430 separate from the MEMS devices 421, 422.

In some embodiments, each device in the second unit 494 is independently addressable via a corresponding local controller 429, such that the operation of each device in the second unit 494 can be controlled without changing the operating status of any other device of showerhead 400B. In some embodiments, each device in the second unit 494 is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers 429. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead 400B that is not within the defined group.

In some embodiments, the third unit 496 includes a local controller, such as local controller 429 of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS device 423 and/or a separate heater, if present) in the third unit 496. In an example, the local controller is integrated into

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the MEMS device 423. In another example, the local controller is coupled to the PCB 430 separate from the MEMS device 423.

In some embodiments, each device in the third unit 496 is independently addressable via a corresponding local controller 429, such that the operation of each device in the third unit 496 can be controlled without changing the operating status of any other device of showerhead 400B. In some embodiments, each device in the third unit 496 is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers 429. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead 400B that is not within the defined group.

In some embodiments, the second unit 484 of FIG. 4A and the second unit 494 of FIG. 4B may be configured to provide independent control of a first gas flow through the first port 432 of the PCB 430 and a second gas flow through the second port 434 of the PCB 430. For example, an operator can configure the second unit 484 of FIG. 4A or the second unit 494 of FIG. 4B to operate under any one of four modes: (i) flow gas from the first conduit 442 of the manifold 440 only; (ii) flow gas from the second conduit 444 of the manifold 440 only; (iii) flow gas from both the first conduit 442 and the second conduit 444 of the manifold 440; and (iv) do not flow gas from the manifold 440.

In the first mode, valve member 426 of MEMS device 421 is positioned to at least partially uncover orifice 425, and valve member 428 of MEMS device 422 is positioned to obscure orifice 427. In the second mode, valve member 426 of MEMS device 421 is positioned to obscure orifice 425, and valve member 428 of MEMS device 422 is positioned to at least partially uncover orifice 427. In the third mode, valve member 426 of MEMS device 421 is positioned to at least partially uncover orifice 425, and valve member 428 of MEMS device 422 is positioned to at least partially uncover orifice 427. In the fourth mode, valve member 426 of MEMS device 421 is positioned to obscure orifice 425, and valve member 428 of MEMS device 422 is positioned to obscure orifice 427.

It is contemplated that an operator can configure the second unit 484 of FIG. 4A or the second unit 494 of FIG. 4B to switch from one of the first, second, third, or fourth modes to another of the first, second, third, or fourth modes. In an example, an operator may control the second unit 484 of FIG. 4A or the second unit 494 of FIG. 4B to switch between modes in order to adjust a composition of process gases within a zone of a processing volume (120, FIG. 1) of a processing chamber (100, FIG. 1).

In some embodiments, the third unit 486 of FIG. 4A and the third unit 496 of FIG. 4B may be configured to provide independent control of a first gas flow through the first port 432 of the PCB 430 and a second gas flow through the second port 434 of the PCB 430. For example, an operator can configure the third unit 486 of FIG. 4A or the third unit 496 of FIG. 4B to operate under any one of four modes: (i) flow gas from the first conduit 442 of the manifold 440 only; (ii) flow gas from the second conduit 444 of the manifold 440 only; (iii) flow gas from both the first conduit 442 and the second conduit 444 of the manifold 440; and (iv) do not flow gas from the manifold 440.

In the first mode, first valve member 454 of MEMS device 423 is positioned to at least partially uncover first orifice 453, and second valve member 456 of MEMS device 423 is positioned to obscure second orifice 455. In the second mode, first valve member 454 of MEMS device 423 is

positioned to obscure first orifice **453**, and second valve member **456** of MEMS device **423** is positioned to at least partially uncover second orifice **455**. In the third mode, first valve member **454** of MEMS device **423** is positioned to at least partially uncover first orifice **453**, and second valve member **456** of MEMS device **423** is positioned to at least partially uncover second orifice **455**. In the fourth mode, first valve member **454** of MEMS device **423** is positioned to obscure first orifice **453**, and second valve member **456** of MEMS device **423** is positioned to obscure second orifice **455**.

It is contemplated that an operator can configure the third unit **486** of FIG. **4A** or the third unit **496** of FIG. **4B** to switch from one of the first, second, third, or fourth modes to another of the first, second, third, or fourth modes. In an example, an operator may control the third unit **486** of FIG. **4A** or the third unit **496** of FIG. **4B** to switch between modes in order to adjust a composition of process gases within a zone of a processing volume (**120**, FIG. **1**) of a processing chamber (**100**, FIG. **1**).

The configuration of manifold **440** and PCB **430** depicted in FIGS. **4A** and **4B** facilitates expedient switching of gases that are flowed through the showerheads **400A** and **400B**. In an example, a first processing operation involves flowing a first gas through first conduit **442**, first ducts **446**, first ports **432**, one or more MEMS devices **420**, **421**, **423**, and through openings **418** of the faceplate **410** into a processing volume of a processing chamber. A second processing operation involves flowing a different second gas through first conduit **442**, first ducts **446**, first ports **432**, one or more MEMS devices **420**, **421**, **423**, and through openings **418** of the faceplate **410** into the processing volume of the processing chamber. In the event that it is detrimental or undesirable to mix the first gas and the second gas, then the first gas must be purged out of the first conduit **442**, the first ducts **446**, the first ports **432**, the one or more MEMS devices **420**, **421**, **423**, and the processing chamber before flowing the second gas. Because the total volume of the first conduit **442** and the first ducts **446** of the manifold plus the first ports **432** of the PCB **430** is less than the volume of a plenum, such as plenum **116**, the quantity of first gas that must be purged and potentially wasted is less for a processing chamber incorporating showerhead **400A** or **400B** than for a processing chamber that delivers process gases via a plenum. Additionally, the time required for the purging operation is less for a processing chamber incorporating showerhead **400A** or **400B** than for a processing chamber that delivers process gases via a plenum. Consequently, a processing chamber incorporating showerhead **400A** or **400B** provides operational efficiencies of time, gas wastage, throughput, and cost over a processing chamber that delivers process gases via a plenum.

The configuration of manifold **440** and PCB **430** depicted in FIGS. **4A** and **4B** facilitates expedient simultaneous delivery of gases that are flowed through the showerheads **400A** and **400B**. In an example, a first gas is flowed through first conduit **442**, first ducts **446**, and one or more MEMS devices **421**, **423**. A second gas is flowed through second conduit **444**, second ducts **448**, and one or more MEMS devices **422**, **423**. The first and second gases are mixed in the void spaces **474** of the second units **484**, **494** and third units **486**, **496** before passing through the openings **418** of faceplate **410**. The MEMS devices **421**, **422**, **423** facilitate localized adjustments to the ratio of first and second gases within each unit **484**, **486**, **494**, **496**. Furthermore, the MEMS devices **421**, **422**, **423** facilitate localized adjust-

ments to the combined flowrate of first and second gases within each unit **484**, **486**, **494**, **496**.

Consequently, the flowrates and relative quantities of the first and second gas within different regions of the processing volume of a processing chamber can be tailored. In some embodiments, the flowrates and relative quantities of the first and second gas are adjusted towards being uniform across the processing volume. In some embodiments, the flowrates and relative quantities of the first and second gas are adjusted to provide a greater flow of gas in a first region of the processing volume and a lesser flow of gas in a second region of the processing volume. In some embodiments, the flowrates and relative quantities of the first and second gas are adjusted to provide a greater proportion of the first gas than the second gas in a first region of the processing volume, and a greater proportion of the second gas than the first gas in a second region of the processing volume.

FIG. **5A** is a schematic cross-sectional side view of an exemplary showerhead **500A**. It is contemplated that the configuration of showerhead **500A** may be used as showerhead **112** of FIG. **1**. Showerhead **500A** includes a faceplate **510A** formed of multiple MEMS modules **520** that are suspended from a PCB **530A**. It is contemplated that the PCB **530A** may be configured similarly to PCB **330** of FIG. **3**. As illustrated, in some embodiments, a portion of the PCB **530A** may extend through the chamber body **102** of the processing chamber. Such a configuration facilitates connection to a master controller **550** external to the processing chamber.

Each MEMS module **520** includes a body **521** with sidewalls **522** containing a MEMS device **560**. In some embodiments, the body **521** is integrated with MEMS device **560**. In some embodiments, the body **521** may be attached to MEMS device **560**. In another embodiment, the body **521** is attached to the PCB **530A** separately from the MEMS device **560**. The sidewalls **522** extend below MEMS device **560** to a base **524**. One or more holes **526** in the base **524** facilitate gases to flow through the MEMS module **560** and into the processing volume (**120**, FIG. **1**) of a processing chamber, such as processing chamber **100**. In some embodiments, a diffuser **528** is disposed above the one or more holes. The diffuser **528** facilitates a uniform distribution of gas through the one or more holes **526**. In some embodiments, the diffuser **528** filters out particles entrained in the gas. Example diffusers **528** include a mesh (such as a sintered mesh), a porous metal filter, a foam (such as porous PTFE foam), or the like. In some embodiments, the diffuser **528** may be omitted.

To inhibit corrosion, it is contemplated that each MEMS module **520** may be manufactured out of a corrosion-resistant material, such as a ceramic or a metal, such as titanium. Additionally, or alternatively, surfaces of each MEMS module **520** may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as Al_2O_3), yttrium oxides (such as Y_2O_3), silicon oxides (such as SiO_2), titanium oxides (such as TiO_2), and the like.

Each MEMS device **560** associated with a corresponding MEMS module **520** may be configured similarly to MEMS device **200**. Each MEMS device **560** is schematically depicted to include an orifice **572**, a valve member **574**, a heater **576**, and a sensor **578**, such as described above for MEMS device **200**. Each MEMS device **560** is coupled to the PCB **530A**. In some embodiments, the MEMS device **560** is soldered to the PCB **530A**. In some of such embodi-

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ments, the solder surrounds the orifice 572 and provides a seal between the PCB 530A and the MEMS device 560. Each contact of each MEMS device 560 is connected to the PCB 530A. The sensor 578, the heater 576, and the valve member 574 of each MEMS device 560 receive electrical power via the PCB 530A. In some embodiments, sensor 578 is omitted, and a sensor separate from the MEMS device 560 is coupled to the PCB 530A. Such a sensor may measure any one or more of pressure, temperature, or flow rate. In some embodiments, heater 576 is omitted, and a heater separate from the MEMS device 560 is coupled to the PCB 530A. The PCB 530A is coupled to a master controller 550 for the transmission of power and/or control signals and/or telemetry with each MEMS device 560.

The PCB 530A includes a port 531 associated with each MEMS device 560. When the valve member 574 of a MEMS device 560 permits gas to flow through the corresponding orifice 572, gas in the plenum 116 can flow through the corresponding port 531 in the PCB 530A and through the orifice 572 into the corresponding MEMS module 520. In some embodiments, the gas is heated by the heater 576. The gas then flows through the one or more holes 526 in the base 524 of the MEMS module 520.

In some embodiments, the MEMS module 520 includes a local controller, such as local controller 429 of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS device 560 and/or a separate heater, if present) in the MEMS module 520. In an example, the local controller is integrated into the MEMS module 520, such as by being integrated into the MEMS device 560. In another example, the local controller is coupled to the PCB 530A separate from the MEMS device 520.

In some embodiments, each device (such as MEMS device 560 and/or a separate heater) in the MEMS module 520 is independently addressable via a corresponding local controller, such as local controller 329, 429. The operation of each device in the MEMS module 520 can be controlled without changing the operating status of any other device of showerhead 500A. In some embodiments, each device in the MEMS module 520 is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead 500A that is not within the defined group.

It is contemplated that clusters of devices and/or clusters of MEMS modules 520 of showerhead 500A may be controlled in similar ways to the examples provided above with respect to clusters of MEMS devices 320 of showerhead 300.

FIG. 5B is a schematic cross-sectional side view of an exemplary showerhead 500B, which is a variant of showerhead 500A, and incorporates elements of showerhead 400B. It is contemplated that the configuration of showerhead 500B may be used as showerhead 112 of FIG. 1. Showerhead 500B includes a faceplate 510B formed of multiple MEMS modules 520, 520A, 520B that are suspended from a PCB 530B. It is contemplated that the PCB 530B may be configured similarly to PCB 430 of FIGS. 4A and 4B. Showerhead 500B includes a manifold 540 disposed on the PCB 530B. The manifold 540 includes a first conduit 542 for passage of a first gas, and a second conduit 544 for passage of a second gas. It is contemplated that the first conduit 542 is isolated from the second conduit 544 such that the first gas and the second gas do not mix in the

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manifold 540. First ducts 546 from the first conduit 542 are aligned with first ports 532 in the PCB 530B. Second ducts 548 from the second conduit 544 are aligned with second ports 534 in the PCB 530B. An interface 538 between the manifold 540 and the PCB 530B is sealed, such as by bonding the manifold 540 to the PCB 530B, to inhibit mixing of the first gas and the second gas at the interface 538. The PCB 530B is coupled to master controller 550 for the transmission of power and/or control signals and/or telemetry with MEMS devices attached to the PCB 530B. As illustrated, in some embodiments, a portion of the PCB 530B may extend through the chamber body 102 of the processing chamber. Such a configuration facilitates connection to master controller 550.

In some embodiments, manifold 540 includes one or more additional conduits and corresponding ducts configured to convey one or more additional gases. In such embodiments, the one or more additional conduits may be isolated from the first conduit 542 and the second conduit 544. Furthermore, it is contemplated that the PCB 530B may include additional ports aligned with the additional ducts. As illustrated, in some embodiments, a portion of the manifold 540 may extend through the chamber body 102 of the processing chamber. Such a configuration facilitates connection to one or more gas supplies.

It is contemplated that clusters of devices and/or clusters of MEMS modules 520, 520A, 520B of showerhead 500B may be controlled in similar ways to the examples provided above with respect to clusters of MEMS devices 320 of showerhead 300.

FIG. 5B illustrates three exemplary configurations in which one or more MEMS modules 520, 520A, 520B are attached to the PCB 530B. Each attachment of a MEMS module 520, 520A, 520B with a corresponding portion of the PCB 530B may be considered as a discrete unit; each unit is described below.

A first unit 582 includes MEMS module 520 (as described above), a second unit 584 includes MEMS module 520A, and a third unit 586 includes MEMS module 520B. MEMS module 520A includes a body 521A, sidewalls 522A, and a base 524A with one or more holes 526A, as described above for MEMS module 520. MEMS module 520B includes a body 521B, sidewalls 522B, and a base 524B with one or more holes 526B, as described above for MEMS module 520.

To inhibit corrosion, it is contemplated that each MEMS module 520, 520A, 520B may be manufactured out of a corrosion-resistant material, such as a ceramic or a metal such as titanium. Additionally, or alternatively, surfaces of each MEMS module 520, 520A, 520B may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as Al_2O_3), yttrium oxides (such as Y_2O_3), silicon oxides (such as SiO_2), titanium oxides (such as TiO_2), and the like.

In the first unit 582, the PCB 530B includes a port 532 aligned with a first duct 546 of the manifold 540, but does not include a port corresponding to a second duct 548 of the manifold 540. Consequently, the first unit 582 is configured to manage gas supplied via the first conduit 542 of the manifold, but is not configured to manage gas supplied via the second conduit 544 of the manifold 540. However, in alternative embodiments, the first unit 582 may be configured to manage gas supplied via the second conduit 544 of the manifold 540, but not configured to manage gas supplied via the first conduit 542 of the manifold 540. In such

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embodiments, the PCB 530B includes a port aligned with a second duct 548 of the manifold 540, but does not include a port corresponding to a first duct 546 of the manifold 540.

MEMS module 520, as described above, is attached to the PCB 530B. A MEMS device 560 associated with MEMS module 520 coupled to the PCB 530B regulates gas flow through the port 532 of the PCB 530B. It is contemplated that the MEMS device 560 may be configured similarly to MEMS device 200. The MEMS device 560 is schematically depicted to include an orifice 572, a valve member 574, a heater 576, and a sensor 578, such as described above for MEMS device 200. The MEMS device 560 is coupled to the PCB 530B. In some embodiments, the MEMS device 560 is soldered to the PCB 530B. In some of such embodiments, the solder surrounds the orifice 572 and provides a seal between the PCB 530B and the MEMS device 560. Each contact of the MEMS device 560 is connected to the PCB. The sensor 578, the heater 576, and the valve member 574 of the MEMS device 560 receive electrical power via the PCB 530B.

In some embodiments, sensor 578 is omitted, and a sensor separate from the MEMS device 560 is coupled to the PCB 530B. Such a sensor may measure any one or more of pressure, temperature, or flow rate. In some embodiments, heater 576 is omitted, and a heater separate from the MEMS device 560 is coupled to the PCB 530B. The PCB 530B is coupled to a master controller 550 for the transmission of power and/or control signals and/or telemetry with the MEMS device 560.

In some embodiments, the MEMS module 520 includes a diffuser, such as diffuser 528 described above. In some embodiments, the diffuser may be omitted.

In some embodiments, the first unit 582 includes a local controller, such as local controller 429 of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS device 560 and/or a separate heater, if present) in the first unit 582. In an example, the local controller is integrated into the MEMS module 520, such as by being integrated into the MEMS device 560. In another example, the local controller is coupled to the PCB 530B separate from the MEMS device 560.

In some embodiments, each device in the first unit 582 is independently addressable via a corresponding local controller (such as local controller 429), such that the operation of each device in the first unit 582 can be controlled without changing the operating status of any other device of showerhead 500B. In some embodiments, each device in the first unit 582 is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead 500B that is not within the defined group.

In the second unit 584, the PCB 530B includes a first port 532 aligned with a first duct 546 of the manifold 540, and a second port 534 aligned with a second duct 544 of the manifold 540. Consequently, the second unit 584 is configured to manage gas supplied via the first conduit 542 of the manifold 540 and via the second conduit 544 of the manifold 540.

MEMS module 520A is attached to the PCB 530B. MEMS module 520A includes a first MEMS device 562 coupled to the PCB 530B that regulates gas flow through the first port 532 of the PCB 530B. MEMS module 520A includes a second MEMS device 564 coupled to the PCB

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530B that regulates gas flow through the second port 534 of the PCB 530B. It is contemplated that each MEMS device 562, 564 may be configured similarly to MEMS device 200.

Each MEMS device 562, 564 is schematically depicted to include, respectively, an orifice 572A, 572B; a valve member 574A, 574B; and a heater 576A, 576B; such as described above for MEMS device 200. In some embodiments, heaters 576A, 576B may be omitted, and a heater separate from each MEMS device 562, 564 is coupled to the PCB 530B. As illustrated, in some embodiments, at least one of the MEMS devices 562, 564 includes a sensor 578A, such as sensor 236. In some embodiments, sensor 578A is omitted, and a sensor separate from the MEMS devices 562, 564 is coupled to the PCB 530B. Such a sensor may measure any one or more of pressure, temperature, or flow rate. Each MEMS device 562, 564 is coupled to the PCB 530B. In some embodiments, each MEMS device 562, 564 is soldered to the PCB 530B. In some of such embodiments, the solder surrounds each orifice 572A, 572B and provides a seal between the PCB 530B and each MEMS device 562, 564. Each contact of each MEMS device 562, 564 is connected to the PCB 530B. The sensor 578A; the heater 576A, 576B (including a separate heater when present); and the valve member 574A, 574B of each MEMS device 562, 564 receive electrical power via the PCB 530B.

In some embodiments, the MEMS module 520A includes a diffuser, such as diffuser 528 described above. In some embodiments, the diffuser may be omitted.

In some embodiments, the second unit 584 includes a local controller, such as local controller 429 of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS devices 562, 564 and/or a separate heater, if present) in the second unit 584. In an example, the local controller is integrated into one of the MEMS devices 562, 564. In another example, the local controller is coupled to the PCB 530B separate from the MEMS devices 562, 564.

In some embodiments, each device in the second unit 584 is independently addressable via a corresponding local controller (such as local controller 429), such that the operation of each device in the second unit 584 can be controlled without changing the operating status of any other device of showerhead 500B. In some embodiments, each device in the second unit 584 is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead 500B that is not within the defined group.

In some embodiments, the second unit 584 may be configured to provide independent control of a first gas flow through the first port 532 of the PCB 530B and a second gas flow through the second port 534 of the PCB 530B. For example, an operator can configure the second unit 584 to operate under any one of four modes: (i) flow gas from the first conduit 542 of the manifold 540 only; (ii) flow gas from the second conduit 544 of the manifold 540 only; (iii) flow gas from both the first conduit 542 and the second conduit 544 of the manifold 540; and (iv) do not flow gas from the manifold 540.

In the first mode, valve member 574A of MEMS device 562 is positioned to at least partially uncover orifice 572A, and valve member 574B of MEMS device 564 is positioned to obscure orifice 572B. In the second mode, valve member 574A of MEMS device 562 is positioned to obscure orifice 572A, and valve member 574B of MEMS device 564 is

positioned to at least partially uncover orifice 572B. In the third mode, valve member 574A of MEMS device 562 is positioned to at least partially uncover orifice 572A, and valve member 574B of MEMS device 564 is positioned to at least partially uncover orifice 572B. In the fourth mode, 5 574A of MEMS device 562 is positioned to obscure orifice 572A, and valve member 574B of MEMS device 564 is positioned to obscure orifice 572B.

It is contemplated that an operator can configure the second unit 584 to switch from one of the first, second, third, or fourth modes to another of the first, second, third, or fourth modes. In an example, an operator may control the second unit 584 to switch between modes in order to adjust a composition of process gases within a zone of a processing volume (120, FIG. 1) of a processing chamber (100, FIG. 1).

In the third unit 586, the PCB 530B includes a first port 532 aligned with a first duct 546 of the manifold 540, and a second port 534 aligned with a second duct 548 of the manifold 540. Consequently, the third unit 586 is configured 20 to manage gas supplied via the first conduit 542 of the manifold 540 and via the second conduit 542 of the manifold 540.

MEMS module 520B is attached to the PCB 530B. MEMS module 520B includes a MEMS device 566 coupled to the PCB 530B that regulates gas flow through the first port 532 of the PCB 530B and regulates gas flow through the second port 534 of the PCB 530B. It is contemplated that the MEMS device 566 may be configured similarly to MEMS device 250.

The MEMS device 566 is schematically depicted to include a first orifice 572C and a first valve member 574C for controlling gas flow through the first port 532 of the PCB 530B, such as described above for MEMS device 250. The MEMS device 566 is schematically depicted to include a second orifice 572D and a second valve member 574D for controlling gas flow through the second port 534 of the PCB 530B, such as described above for MEMS device 250. In some embodiments, the MEMS device 566 is soldered to the PCB 530B. In some of such embodiments, the solder surrounds the first orifice 572C and the second orifice 572D, and provides a seal between the PCB 530B and the MEMS device 566. The MEMS device 566 is schematically depicted to include a first heater 576C associated with the first orifice 572C and a second heater 576D associated with the second orifice 572D, such as described above for MEMS device 250. In some embodiments, heaters 576C, 576D are omitted, and a heater separate from the MEMS device 566 is coupled to the PCB 530B. The MEMS device 566 is schematically depicted to include a sensor 578B, such as described above for MEMS device 250. In some embodiments, sensor 578B is omitted, and a sensor separate from the MEMS device 566 is coupled to the PCB 530B. Such a sensor may measure any one or more of pressure, temperature, or flow rate. The sensor 578B, the heaters 576C, 576D, and the valve members 574C, 574D of the MEMS device 566, and/or a separate heater (if present) receive electrical power via the PCB 530B.

In some embodiments, the MEMS module 520B includes a diffuser, such as diffuser 528 described above. In some embodiments, the diffuser may be omitted.

In some embodiments, the third unit 586 includes a local controller, such as local controller 429 of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of at least one device (such as the MEMS device 566 and/or a separate heater, if present) in the third unit 586. In an example, the local controller is integrated into

the MEMS device 566. In another example, the local controller is coupled to the PCB 530B separate from the MEMS device 566.

In some embodiments, each device in the third unit 586 is independently addressable via a corresponding local controller (such as local controller 429), such that the operation of each device in the third unit 586 can be controlled without changing the operating status of any other device of showerhead 500B. In some embodiments, each device in the third unit 586 is assigned to one or more groups of devices, and each group of devices is independently addressable via one or more corresponding local controllers. In such embodiments, the operation of each device within a defined group can be controlled without changing the operating status of any other device of showerhead 500B that is not within the defined group.

In some embodiments, the third unit 586 may be configured to provide independent control of a first gas flow through the first port 532 of the PCB 530B and a second gas flow through the second port 534 of the PCB 530B. For example, an operator can configure the third unit 586 to operate under any one of four modes: (i) flow gas from the first conduit 542 of the manifold 540 only; (ii) flow gas from the second conduit 544 of the manifold 540 only; (iii) flow gas from both the first conduit 542 and the second conduit 544 of the manifold 540; and (iv) do not flow gas from the manifold 540.

In the first mode, first valve member 574C of MEMS device 566 is positioned to at least partially uncover first orifice 572C, and second valve member 574D of MEMS device 566 is positioned to obscure second orifice 572D. In the second mode, first valve member 574C of MEMS device 566 is positioned to obscure first orifice 572C, and second valve member 574D of MEMS device 566 is positioned to at least partially uncover second orifice 572D. In the third mode, first valve member 574C of MEMS device 566 is positioned to at least partially uncover first orifice 572C, and second valve member 574D of MEMS device 566 is positioned to at least partially uncover second orifice 572D. In the fourth mode, first valve member 574C of MEMS device 566 is positioned to obscure first orifice 572C, and second valve member 574D of MEMS device 566 is positioned to obscure second orifice 572D.

It is contemplated that an operator can configure the third unit 586 to switch from one of the first, second, third, or fourth modes to another of the first, second, third, or fourth modes. In an example, an operator may control the third unit 586 to switch between modes in order to adjust a composition of process gases within a zone of a processing volume (120, FIG. 1) of a processing chamber (100, FIG. 1).

The configuration of manifold 540 and PCB 530B depicted in FIG. 5B facilitates expedient switching of gases that are flowed through showerhead 500B. In an example, a first processing operation involves flowing a first gas through first conduit 542, first ducts 546, first ports 532, one or more MEMS devices 560, 562, 566, and through the one or more holes 526, 526A, 526B of the faceplate 510B into a processing volume of a processing chamber. A second processing operation involves flowing a different second gas through first conduit 542, first ducts 546, first ports 532, one or more MEMS devices 560, 562, 566, and through the one or more holes 526, 526A, 526B of the faceplate 510B into the processing volume of the processing chamber. In the event that it is detrimental or undesirable to mix the first gas and the second gas, then the first gas must be purged out of the first conduit 542, the first ducts 546, the first ports 532, the one or more MEMS devices 560, 562, 566, and the

processing chamber before flowing the second gas. Because the total volume of the first conduit **442** and the first ducts **446** of the manifold plus the first ports **532** of the PCB **530B** is less than the volume of a plenum, such as plenum **116**, the quantity of first gas that must be purged and potentially wasted is less for a processing chamber incorporating showerhead **500 B** than for a processing chamber that delivers process gases via a plenum. Additionally, the time required for the purging operation is less for a processing chamber incorporating showerhead **500 B** than for a processing chamber that delivers process gases via a plenum. Consequently, a processing chamber incorporating showerhead **500 B** provides operational efficiencies of time, gas wastage, throughput, and cost over a processing chamber that delivers process gases via a plenum.

The configuration of manifold **540** and PCB **530B** depicted in FIG. **5B** facilitates expedient simultaneous delivery of gases that are flowed through showerhead **500B**. In an example, a first gas is flowed through first conduit **542**, first ducts **546**, first ports **532**, and one or more MEMS devices **562**, **566**. A second gas is flowed through second conduit **544**, second ducts **548**, second ports **534**, and one or more MEMS devices **564**, **566**. The first and second gases are mixed in the MEMS modules **520A**, **520B** of the second **584** and third **586** units, respectively, before passing through the one or more holes **526A**, **526B**, respectively, of the faceplate **510B**. The MEMS devices **562**, **564**, **566** facilitate localized adjustments to the ratio of first and second gases within each unit **584**, **586**. Furthermore, the MEMS devices **562**, **564**, **566** facilitate localized adjustments to the combined flowrate of first and second gases within each unit **584**, **586**.

Consequently, the flowrates and relative quantities of the first and second gas within different regions of the processing volume of a processing chamber can be tailored. In some embodiments, the flowrates and relative quantities of the first and second gas are adjusted towards being uniform across the processing volume. In some embodiments, the flowrates and relative quantities of the first and second gas are adjusted to provide a greater flow of gas in a first region of the processing volume and a lesser flow of gas in a second region of the processing volume. In some embodiments, the flowrates and relative quantities of the first and second gas are adjusted to provide a greater proportion of the first gas than the second gas in a first region of the processing volume, and a greater proportion of the second gas than the first gas in a second region of the processing volume.

FIGS. **6A-6G** are schematic cross-sectional side views of different configurations of MEMS modules. In some embodiments, it is contemplated that showerhead **500A** and showerhead **500B** may include MEMS modules of the same configuration. Additionally, or alternatively, it is contemplated that showerhead **500A** and showerhead **500B** may include MEMS modules of different configurations.

FIG. **6A** illustrates MEMS module **600A**. It is contemplated that MEMS module **600A** may be used as any of MEMS modules **520**, **520A**, or **520B**. MEMS module **600A** is shown as including MEMS device **610A**. It is contemplated that MEMS device **610A** may be configured similarly to MEMS device **200** or MEMS device **250**. MEMS device **610A** is schematically depicted to include an orifice **612A**, a valve member **614A**, a heater **616A** at the orifice **612A**, and a sensor **618A**, such as described above for MEMS device **200**. In some embodiments, it is contemplated that MEMS module **600A** may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module **600A** may contain more than one MEMS device.

MEMS module **600A** includes a body **601A** with sidewalls **602A**. In some embodiments, the body **601A** is integrated with MEMS device **610A**. In some embodiments, the body **601A** may be attached to MEMS device **610A**. In another embodiment, the body **601A** is configured to be attached to a PCB separately from MEMS device **610A**. The sidewalls **602A** extend below MEMS device **610A** to a base **604A**. One or more holes **606A** in the base **604A** facilitate gases to flow through the MEMS device **610A**. The base **604A** is configured as an insert that is coupled to the sidewalls **602A**. In some embodiments, the insert is removable to facilitate, for example, replacement with an alternative insert having different hole sizes, a different number of holes, or a different pattern of holes. In an example, the insert is removed to facilitate cleaning and/or refurbishment of the showerhead. In an example, the insert is coupled to the sidewalls **602A** by a screw thread. In some embodiments, a diffuser, such as diffuser **470**, is disposed above the one or more holes **606A**.

In some embodiments, MEMS module **600A** includes a local controller, such as local controller **429** of FIG. **4A**. In such embodiments, it is contemplated that the local controller controls operation of MEMS device **610A**. In an example, the local controller is integrated into the MEMS device **610A**. In another example, the local controller is coupled to a PCB separately from MEMS device **610A**.

FIG. **6B** illustrates MEMS module **600B**. It is contemplated that MEMS module **600B** may be used as any of MEMS modules **520**, **520A**, or **520B**. MEMS module **600B** is shown as including MEMS device **610B**. It is contemplated that MEMS device **610B** may be configured similarly to MEMS device **200** or MEMS device **250**. MEMS device **610B** is schematically depicted to include an orifice **612B**, a valve member **614B**, a heater **616B** at the orifice **612B**, and a sensor **618B**, such as described above for MEMS device **200**. In some embodiments, it is contemplated that MEMS module **600B** may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module **600B** may contain more than one MEMS device.

MEMS module **600B** includes a body **601B**. In some embodiments, the body **601B** is integrated with MEMS device **610B**. In some embodiments, the body **601B** may be attached to MEMS device **610B**. In another embodiment, the body **601B** is configured to be attached to a PCB separately from MEMS device **610B**. An insert holder **607** is attached to the body **601B**, such as by a weld or an adhesive bond. The insert holder **607** includes sidewalls **602B** extending below MEMS device **610B** to a base **604B**. One or more holes **606B** in the base **604B** facilitate gases to flow through the MEMS device **610B**. The base **604B** is configured as an insert that is coupled to the sidewalls **602B**. In some embodiments, the insert is removable to facilitate, for example, replacement with an alternative insert having different hole sizes, a different number of holes, or a different pattern of holes. In an example, the insert is removed to facilitate cleaning and/or refurbishment of the showerhead. In an example, the insert is coupled to the sidewalls **602B** by a screw thread. In some embodiments, a diffuser, such as diffuser **470**, is disposed above the one or more holes **606B**.

In some embodiments, MEMS module **600B** includes a local controller, such as local controller **429** of FIG. **4A**. In such embodiments, it is contemplated that the local controller controls operation of MEMS device **610B**. In an example, the local controller is integrated into the MEMS device **610B**. In another example, the local controller is coupled to a PCB separately from MEMS device **610B**.

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FIG. 6C illustrates MEMS module 600C. It is contemplated that MEMS module 600C may be used as any of MEMS modules 520, 520A, or 520B. MEMS module 600C is shown as including MEMS device 610C. It is contemplated that MEMS device 610C may be configured similarly to MEMS device 200 or MEMS device 250. MEMS device 610C is schematically depicted to include an orifice 612C, a valve member 614C, a heater 616C at the orifice 612C, and a sensor 618C, such as described above for MEMS device 200. In some embodiments, it is contemplated that MEMS module 600C may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module 600C may contain more than one MEMS device.

MEMS module 600C includes a body 601C. In some embodiments, the body 601C is integrated with MEMS device 610C. In some embodiments, the body 601C may be attached to MEMS device 610C. In another embodiment, the body 601C is configured to be attached to a PCB separately from MEMS device 610C. A base unit 608 is attached to the body 601C, such as by a weld or an adhesive bond. The base unit 608 includes sidewalls 602C extending below MEMS device 610C to a base 604C. One or more holes 606C in the base 604C facilitate gases to flow through the MEMS device 610C. The base 604B is affixed to, or is integral with, the sidewalls 602C. In some embodiments, a diffuser, such as diffuser 470, is disposed above the one or more holes 606C.

In some embodiments, MEMS module 600C includes a local controller, such as local controller 429 of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of MEMS device 610C. In an example, the local controller is integrated into the MEMS device 610C. In another example, the local controller is coupled to a PCB separately from MEMS device 610C.

FIG. 6D illustrates MEMS module 600D. It is contemplated that MEMS module 600D may be used as any of MEMS modules 520, 520A, or 520B. MEMS module 600D is shown as including MEMS device 610D. It is contemplated that MEMS device 610D may be configured similarly to MEMS device 200 or MEMS device 250. MEMS device 610D is schematically depicted to include an orifice 612D, a valve member 614D, and a sensor 618D, such as described above for MEMS device 200. A heater 622 is attached to MEMS device 610D, and extends below MEMS device 610D. In some embodiments, the heater 622 includes a mesh, such as a sintered mesh. In some embodiments, the heater 622 also serves as a diffuser or a filter. In some embodiments, it is contemplated that MEMS module 600D may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module 600D may contain more than one MEMS device.

MEMS module 600D includes a body 601D with sidewalls 602D. In some embodiments, the body 601D is integrated with MEMS device 610D. In some embodiments, the body 601D may be attached to MEMS device 610D. In another embodiment, the body 601D is configured to be attached to a PCB separately from MEMS device 610D. The sidewalls 602D extend below MEMS device 610D to a base 604D. One or more holes 606D in the base 604D facilitate gases to flow through the MEMS device 610D. The base 604D is configured as an insert that is coupled to the sidewalls 602D. In some embodiments, the insert is removable to facilitate, for example, replacement with an alternative insert having different hole sizes, a different number of holes, or a different pattern of holes. In an example, the insert is removed to facilitate cleaning and/or refurbishment

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of the showerhead. In an example, the insert is coupled to the sidewalls 602D by a screw thread. In some embodiments, a diffuser, such as diffuser 470, is disposed above the one or more holes 606D.

In some embodiments, MEMS module 600D includes a local controller, such as local controller 429 of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of MEMS device 610D. In an example, the local controller is integrated into the MEMS device 610D. In another example, the local controller is coupled to a PCB separately from MEMS device 610D.

FIG. 6E illustrates MEMS module 600E. It is contemplated that MEMS module 600E may be used as any of MEMS modules 520, 520A, or 520B. MEMS module 600E is shown as including MEMS device 610E. It is contemplated that MEMS device 610E may be configured similarly to MEMS device 200 or MEMS device 250. MEMS device 610E is schematically depicted to include an orifice 612E, a valve member 614E, and a sensor 618E, such as described above for MEMS device 200. In some embodiments, it is contemplated that MEMS module 600E may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module 600E may contain more than one MEMS device.

MEMS module 600E includes a body 601E with sidewalls 602E. In some embodiments, the body 601E is integrated with MEMS device 610E. In some embodiments, the body 601E may be attached to MEMS device 610E. In another embodiment, the body 601E is configured to be attached to a PCB separately from MEMS device 610E. The sidewalls 602E extend below MEMS device 610E to a base 604E. One or more holes 606E in the base 604E facilitate gases to flow through the MEMS device 610E. The base 604E is configured as an insert that is coupled to the sidewalls 602E. In some embodiments, the insert is removable to facilitate, for example, replacement with an alternative insert having different hole sizes, a different number of holes, or a different pattern of holes. In an example, the insert is removed to facilitate cleaning and/or refurbishment of the showerhead. In an example, the insert is coupled to the sidewalls 602E by a screw thread. In some embodiments, a diffuser, such as diffuser 470, is disposed above the one or more holes 606E.

MEMS module 600E includes a heater 624 that is separate from MEMS device 610E. Heater 624 is attached to body 601E. It is contemplated that electrical connections between the heater 624 and a PCB may be facilitated by wiring through the body 601E and/or through MEMS device 610E. As illustrated, it is contemplated that heater 624 may be located below MEMS device 610E. In some embodiments, the heater 624 includes a mesh, such as a sintered mesh. In some embodiments, the heater 624 also serves as a diffuser or a filter.

In some embodiments, MEMS module 600E includes a local controller, such as local controller 429 of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of MEMS device 610E. In an example, the local controller is integrated into the MEMS device 610E. In another example, the local controller is coupled to a PCB separately from MEMS device 610E.

FIG. 6F illustrates MEMS module 600F. It is contemplated that MEMS module 600F may be used as any of MEMS modules 520, 520A, or 520B. MEMS module 600F is shown as including MEMS device 610F. It is contemplated that MEMS device 610F may be configured similarly to MEMS device 200 or MEMS device 250. MEMS device 610F is schematically depicted to include an orifice 612F, a

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valve member **614F**, and a sensor **618F**, such as described above for MEMS device **200**. In some embodiments, it is contemplated that MEMS module **600F** may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module **600F** may contain more than one MEMS device.

MEMS module **600F** includes a body **601F** with sidewalls **602F**. In some embodiments, the body **601F** is integrated with MEMS device **610F**. In some embodiments, the body **601F** may be attached to MEMS device **610F**. In another embodiment, the body **601F** is configured to be attached to a PCB separately from MEMS device **610F**. The sidewalls **602F** extend below MEMS device **610F** to a base **604F**. One or more holes **606F** in the base **604F** facilitate gases to flow through the MEMS device **610F**. The base **604F** is configured as an insert that is coupled to the sidewalls **602F**. In some embodiments, the insert is removable to facilitate, for example, replacement with an alternative insert having different hole sizes, a different number of holes, or a different pattern of holes. In an example, the insert is removed to facilitate cleaning and/or refurbishment of the showerhead. In an example, the insert is coupled to the sidewalls **602F** by a screw thread.

MEMS module **600F** includes a heater **626** that is separate from MEMS device **610F**. Heater **626** is attached to body **601F**. It is contemplated that electrical connections between the heater **626** and a PCB may be facilitated by wiring through the body **601F** and/or through MEMS device **610F**. In some embodiments, the heater **626** includes a mesh, such as a sintered mesh. In some embodiments, the heater **626** also serves as a diffuser or a filter.

As illustrated, it is contemplated that heater **626** may be located below MEMS device **610F** at, or close to, the base **604F**. As illustrated, in some embodiments it is contemplated that a diffuser **628**, such as diffuser **470**, may be located between the heater **626** and the MEMS device **610F**.

In some embodiments, MEMS module **600F** includes a local controller, such as local controller **429** of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of MEMS device **610F**. In an example, the local controller is integrated into the MEMS device **610F**. In another example, the local controller is coupled to a PCB separately from MEMS device **610F**.

FIG. 6G illustrates MEMS module **600G**. It is contemplated that MEMS module **600G** may be used as any of MEMS modules **520**, **520A**, or **520B**. MEMS module **600G** is shown as including MEMS device **610G**. It is contemplated that MEMS device **610G** may be configured similarly to MEMS device **200** or MEMS device **250**. MEMS device **610G** is schematically depicted to include an orifice **612G**, a valve member **614G**, and a sensor **618G**, such as described above for MEMS device **200**. In some embodiments, it is contemplated that MEMS module **600G** may alternatively include any variant of any MEMS device described above. Additionally, it is contemplated that MEMS module **600G** may contain more than one MEMS device.

MEMS module **600G** includes a body **601G** with sidewalls **602G**. In some embodiments, the body **601G** is integrated with MEMS device **610G**. In some embodiments, the body **601G** may be attached to MEMS device **610G**. In another embodiment, the body **601G** is configured to be attached to a PCB separately from MEMS device **610G**. The sidewalls **602G** extend below MEMS device **610G** to a base **604G**. One or more holes **606G** in the base **604G** facilitate gases to flow through the MEMS device **610G**. The base **604G** is configured as an insert that is coupled to the sidewalls **602G**. In some embodiments, the insert is remov-

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able to facilitate, for example, replacement with an alternative insert having different hole sizes, a different number of holes, or a different pattern of holes. In an example, the insert is removed to facilitate cleaning and/or refurbishment of the showerhead. In an example, the insert is coupled to the sidewalls **602G** by a screw thread. As illustrated, in some embodiments, a diffuser **628**, such as diffuser **470**, is disposed above the one or more holes **606G**.

It is contemplated that at least the base **604G** may serve as a heating element. As an example, the base may be manufactured from graphite, such as in the form of IFS-2B. In some embodiments, the sidewalls **602G** are manufactured from a material similar to that of the base **604G**. In some embodiments, the body **601G** is manufactured from a material similar to that of the base **604G**. It is contemplated that any part of the body **601G**, sidewalls **602G**, or base **604G** that is not manufactured to serve as a heating element may be manufactured out of a corrosion-resistant material, such as a ceramic or a metal such as titanium.

In some embodiments, MEMS module **600G** includes a local controller, such as local controller **429** of FIG. 4A. In such embodiments, it is contemplated that the local controller controls operation of MEMS device **610G**. In an example, the local controller is integrated into the MEMS device **610G**. In another example, the local controller is coupled to a PCB separately from MEMS device **610G**.

To inhibit corrosion and/or reduce a probability that a valve member of a MEMS device may stick in position and become inoperable, it is contemplated that surfaces of each MEMS device **610A-610G**, and/or local controller (if present), and/or heater **622**, **624**, **626**, and/or diffuser (if present) may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as Al_2O_3), yttrium oxides (such as Y_2O_3), silicon oxides (such as SiO_x), titanium oxides (such as TiO_2), and the like.

To inhibit corrosion, it is contemplated that each MEMS module **600A-600F** may be manufactured out of a corrosion-resistant material, such as a ceramic or a metal such as titanium. Additionally, or alternatively, surfaces of each MEMS module **600A-600F** and **600G** may be coated with one or more suitable materials. Examples of coating materials include silicon carbide, parylenes, hydrophobic anti-stiction films applied by molecular vapor deposition, ceramics, aluminum oxides (such as Al_2O_3), yttrium oxides (such as Y_2O_3), silicon oxides (such as SiO_x), titanium oxides (such as TiO_2), and the like.

FIG. 7 is a schematic plan view of a faceplate **700** of a showerhead. It is contemplated that faceplate **700** may represent any faceplate of the present disclosure, such as any of faceplates **310**, **410**, **510A**, **510B** in a corresponding showerhead **300**, **400A**, **400B**, **500A**, **500B** that is used as showerhead **112** of processing chamber **100**. Zones **710** are delineated on faceplate **700**, each zone **710** includes one or more openings **712** through the faceplate **700**. In operation, process gas flow through the one or more openings **712** of a specific zone **710** is controlled by one or more MEMS devices associated with the specific zone **710**. In operation, one or more devices, such as MEMS devices and/or heaters, are associated with each zone **710**, and are controlled independently and/or in groups, as described above. In an example, the operation of each MEMS device or heater can be controlled without changing the operating status of another MEMS device or heater associated with faceplate **700**. It is contemplated that the number, size, and distribu-

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tion of zones 710 may be configured according to the type of process to be conducted in processing chamber 100 and/or the specific process gas(es) to be employed.

It is contemplated that any of PCB 330, 430, 530A, 530B may be made from a ceramic material with metal conductors embedded therein. In some embodiments, it is contemplated that PCBs 530A and 530B may include a provision for coupling to one or more support members. In an example, the one or more support members facilitate PCBs 530A and 530B spanning a processing volume in a processing chamber and bearing the weight of components, such as MEMS modules 520, 520A, 520B, that are suspended from PCBs 530A and 530B. In a further example, the manifold 540 acts as a support member for PCB 530B.

It is contemplated that manifold 440 or 540 may be made from a ceramic material. In some embodiments, it is contemplated that manifold 440 or 540 may include a provision for coupling to one or more support members. In an example, the one or more support members facilitate manifold 440 or 540 spanning a processing volume in a processing chamber and bearing the weight of components, such as PCB 430 or 530B, that are suspended from manifold 440 or 540. In a further example, the PCB 530B acts as a support member for manifold 540.

It is contemplated that any of master controllers 350, 450, 550 includes a central processing unit (CPU), a memory containing instructions, and support circuits for the CPU. The master controller 350, 450, 550 is of any form of a general-purpose computer processor that is used in an industrial setting for controlling various chambers and equipment and/or sub-processors thereon or therein.

The memory, or non-transitory computer readable medium, is one or more of a readily available memory such as random access memory (RAM), read only memory (ROM), floppy disk, hard disk, flash drive, or any other form of digital storage, local or remote. The support circuits are coupled to the CPU for supporting the CPU (a processor). The support circuits include cache, power supplies, clock circuits, input/output circuitry and subsystems, and the like. Operations and operating parameters are stored in the memory as a software routine that is executed or invoked to turn the master controller 350, 450, 550 into a specific purpose controller to control the operations of any of showerhead 300, 400A, 400B, 500A, 500B. The master controller 350, 450, 550 is configured to conduct any of the operations described herein. The instructions stored on the memory, when executed, cause one or more of the operations described herein to be conducted.

In some embodiments, data from any of sensors 236, 328, 578, 578A, 578B, 618A, 618B, 618C, 618D, 618E, 618F, 618G and/or any sensor associated with any of showerhead 300, 400A, 400B, 500A, 500B may be used to provide feedback to a controller, such as any of master controller 350, 450, 550. In some embodiments, data of electrical current through any of heaters 326, 424, 576, 576A, 576B, 576C, 576D, 616A, 616B, 616C, 622, 624, 626, and/or any heater associated with any of showerhead 300, 400A, 400B, 500A, 500B may be used to provide feedback to a controller, such as any of master controller 350, 450, 550. For example, the data may include pressure data and/or temperature data.

The controller, such as any of master controller 350, 450, 550 uses the data so provided as an input to process commands addressed to individual MEMS device(s) and/or group(s) of MEMS devices. In some of such embodiments, the commands prompt the individual MEMS device(s) and/or group(s) of MEMS devices to adjust a flowrate of a gas through the individual MEMS device(s) and/or group(s) of

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MEMS devices. In an example, such adjustments provide control of flow distribution of one or more gases through any of showerhead 300, 400A, 400B, 500A, 500B. In some of such embodiments, the commands prompt the individual MEMS device(s) and/or group(s) of MEMS devices to adjust a temperature of a gas flowing through the individual MEMS device(s) and/or group(s) of MEMS devices. In an example, such adjustments provide control of temperature distribution of one or more gases through any of showerhead 300, 400A, 400B, 500A, 500B.

The instructions in the memory of the master controller 350, 450, 550 can include one or more machine learning/artificial intelligence algorithms that can be executed in addition to the operations described herein. As an example, a machine learning/artificial intelligence algorithm executed by the master controller 350, 450, 550 can optimize and alter operational parameters based on one or more sensor measurements taken by the one or more sensors 236, 328, 578, 578A, 578B, 618A-618G. The operational parameters can include, for example, pressure, temperature, gas flow rate, valve member position, and heater status.

The one or more machine learning/artificial intelligence algorithms can account for variations in gas flow rate, gas ratios, temperature, and pressure across any of the showerheads described herein. In some embodiments, the one or more machine learning/artificial intelligence algorithms can measure any of the above parameters to determine whether an appropriate quantity of a process gas at an appropriate pressure and an appropriate temperature is being delivered to the appropriate regions of a processing volume according to a prescribed operation. In some embodiments, the one or more machine learning/artificial intelligence algorithms can prompt the master controller 350, 450, 550 to initiate corrective action in order to adjust the quantity, pressure, or temperature of a process gas being delivered to a prescribed region of a processing volume.

The operational flexibilities described above facilitated by showerheads of the present disclosure are not provided by conventional showerheads. Conventional showerheads typically are configured for achieving an optimal distribution of a specific process gas by having a given number of openings of particular sizes arranged in a specific pattern. However, the number, sizing, and arrangement of openings that provides for an optimal distribution of a one process gas do not necessarily provide for an optimal distribution of a different process gas. In contrast, showerheads of the present disclosure provide for the relative flow rate of any process gas through an opening, or a cluster of openings, to be tailored provide for an optimal distribution of any process gas.

The operational flexibilities facilitated by showerheads of the present disclosure provide time and cost efficiencies compared to conventional operations. For example, some conventional processing sequences involve performing a first operation on a substrate with a first process gas in a first processing chamber, then transferring the substrate to a second processing chamber, then performing a second operation on the substrate with a second, different, process gas. In contrast, processing chambers incorporating showerheads of the present disclosure may be used to perform both the first operation with the first process gas and then the second operation with the second process gas. Thus, the need for the second processing chamber is alleviated, as is the time taken to transfer the substrate between the processing chambers.

In another example, some conventional processing sequences involve performing a first operation on a substrate with a first process gas in a first processing chamber includ-

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ing a first conventional showerhead, then transferring the substrate to a second processing chamber that includes a second conventional showerhead, then performing a second operation on the substrate in the processing chamber with a second, different, process gas. In contrast, processing chambers incorporating showerheads of the present disclosure may be used to perform both the first operation with the first process gas and then the second operation with the second process gas using the same showerhead in the same processing chamber. Thus, the need for transferring the substrate between processing chambers is alleviated, as is the time taken to transfer the substrate.

Furthermore, the operational flexibilities facilitated by showerheads of the present disclosure provide efficiencies in inventory management compared to conventional operations. The use of showerheads of the present disclosure reduces or eliminates the need to stock different showerheads that are configured for use with different process gases for different process operations.

It is contemplated that elements and features of any one disclosed embodiment may be beneficially incorporated in one or more other embodiments. While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

What is claimed is:

1. A showerhead for a processing chamber, the showerhead comprising:

a printed circuit board including a plurality of ports therethrough; and

a faceplate including:

a plurality of MEMS modules coupled to the printed circuit board, each MEMS module including:

a body with sidewalls extending to a base, the base including a plurality of holes; and

a MEMS device disposed in the body, the MEMS device including an orifice and a corresponding valve member operable to control fluid flow from at least one of the plurality of ports to the plurality of holes.

2. The showerhead of claim 1, wherein

the valve member is movable between first and second positions to regulate the fluid flow through the orifice.

3. The showerhead of claim 2, wherein each MEMS module includes a heater configured to heat a fluid conveyed through the orifice.

4. The showerhead of claim 2, wherein the valve member blocks fluid flow through the orifice when in the first position.

5. The showerhead of claim 1, further comprising a local controller coupled to the printed circuit board, the local controller configured to control operation of a corresponding MEMS device of the plurality of MEMS modules.

6. The showerhead of claim 1, further comprising a manifold coupled to the printed circuit board, the manifold including a first conduit and a plurality of first ducts, each first duct associated with a corresponding first port of the plurality of ports of the printed circuit board.

7. The showerhead of claim 6, wherein the manifold includes a second conduit and a plurality of second ducts, each second duct associated with a corresponding second port of the plurality of ports of the printed circuit board.

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8. The showerhead of claim 7, wherein:

the MEMS device includes a first MEMS device and a second MEMS device;

the first MEMS device is associated with a corresponding first port of the plurality of ports; and

the second MEMS device is associated with a corresponding second port of the plurality of ports.

9. The showerhead of claim 1, wherein each MEMS module of the plurality of MEMS modules is suspended from the printed circuit board.

10. The showerhead of claim 1, wherein each MEMS device is suspended from the printed circuit board.

11. The showerhead of claim 1, wherein at least one MEMS module of the plurality of MEMS modules includes a diffuser disposed above the plurality of holes.

12. A showerhead for a processing chamber, the showerhead comprising:

a printed circuit board including a plurality of ports therethrough; and

a faceplate including:

a plurality of MEMS modules suspended from the printed circuit board, each MEMS module including:

a body with sidewalls extending to a base, the base including a plurality of holes; and

a MEMS device disposed in the body and spaced from the plurality of holes, the MEMS device including an orifice and a valve member operable to control fluid flow from at least one of the plurality of ports, through the orifice, and to the plurality of holes.

13. The showerhead of claim 12, wherein:

the valve member is movable between first and second positions to regulate fluid flow through the orifice; and the valve member blocks fluid flow through the orifice when in the first position.

14. The showerhead of claim 12, wherein at least one MEMS module of the plurality of MEMS modules includes a heater configured to heat a fluid conveyed through the orifice.

15. The showerhead of claim 12, wherein at least one MEMS module of the plurality of MEMS modules includes a diffuser disposed above the plurality of holes.

16. The showerhead of claim 12, further comprising a local controller coupled to the printed circuit board, the local controller configured to control operation of a corresponding MEMS device of the plurality of MEMS modules.

17. The showerhead of claim 12, further comprising a manifold coupled to the printed circuit board, the manifold including a first conduit and a plurality of first ducts, each first duct associated with a corresponding first port of the plurality of ports of the printed circuit board.

18. The showerhead of claim 17, wherein the manifold includes a second conduit and a plurality of second ducts, each second duct associated with a corresponding second port of the plurality of ports of the printed circuit board.

19. The showerhead of claim 18, wherein:

the MEMS device includes a first MEMS device and a second MEMS device;

the first MEMS device is associated with a corresponding first port of the plurality of ports; and

the second MEMS device is associated with a corresponding second port of the plurality of ports.

20. A processing chamber comprising:

a chamber body;

a printed circuit board including a plurality of ports therethrough; and

a showerhead disposed in the chamber body, the showerhead comprising:
a faceplate including:
a plurality of MEMS modules coupled to the printed circuit board, each MEMS module including: 5
a body with sidewalls extending to a base, the base including a plurality of holes; and
a MEMS device disposed in the body, the MEMS device including an orifice and a corresponding valve member operable to control fluid flow 10
from at least one of the plurality of ports to the plurality of holes and
a controller coupled to the printed circuit board, and configured to control operations at least one MEMS device of the plurality of MEMS devices indepen- 15
dently of operations of other MEMS devices of the plurality of MEMS devices.

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